

Off-Target Impacts of Targeted Policy:

Evidence from the World’s Largest Anti-Narcotics Intervention^{*}

Raahil Madhok[†]

University of Minnesota

Tatiana Zarate-Barrera[‡]

Texas A&M University

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Abstract

This paper documents agricultural spillovers from the world’s largest place-based policy to combat cocaine trafficking. Aerial fumigation of illicit coca crops with glyphosate was the pillar of Colombia’s anti-drug strategy, until the practice was banned in 2015. Using a novel atmospheric dispersion model to measure wind-borne herbicide drift, along with a two-way fixed effects design, we show that wind drift from spraying illicit coca farms reduces harvests of *legitimate* crops by 13.2%. These spillover damages are persistent and adaptation is minimal in the medium run. Difference-in-differences estimates show that, after the ban on spraying, only a small fraction of agricultural losses were recovered. Cropland that failed to regenerate transitions toward grassland and shrubland. In 2025, Colombia announced plans to reinstate aerial spraying. Our policy simulations show that targeting only the largest coca hotspots could avoid \$USD 639 million in spillover damages, equivalent to 3.7% of Colombia’s agricultural GDP.

Keywords: Spillovers, herbicide spraying, dispersion model, enforcement, agriculture

JEL Codes: Q10, Q56, Q15, O13

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[†]Department of Applied Economics; rmadhok@umn.edu

[‡]Department of International Affairs; tatianazarate@tamu.edu

1 Introduction

Geographic targeting is a common policy to curb undesirable activities by focusing enforcement on specific places rather than on individuals. From protected areas to hotspot policing, the objective is to allocate resources where the marginal impact of enforcement is the highest (Blattman et al., 2021; Englander, 2023). However, targeting specific locations can generate spillovers into neighboring areas (Hanson and Rohlin, 2013, 2021; Vieira et al., 2023). These externalities, whether economic, environmental, or social, often cross jurisdictional boundaries, potentially undermining and even reversing policy objectives on net (Gray and Shimshack, 2011; Neumark and Simpson, 2015).

The tradeoff between direct and spillover effects is particularly relevant in agricultural enforcement, where hazardous chemicals are sprayed to kill unwanted vegetation (Camacho and Mejía, 2017). Although these interventions target specific locations, they can also damage nearby, non-targeted crops. Yet little is known about the magnitude and spatial reach of these spillovers. Filling this gap is important for low- and middle-income countries where rural communities rely heavily on small-scale agriculture for subsistence and income.

This paper examines the off-target effects of the world’s largest place-based drug eradication effort: Colombia’s aerial spraying of coca, the primary input in cocaine production. Backed by U.S. funding under an initiative known as Plan Colombia, national authorities sprayed glyphosate—a widely used and potent herbicide—from aircraft over coca fields for over two decades. While glyphosate kills unwanted vegetation (Benbrook, 2016), it can also cause harmful externalities as wind drift and water runoff carry the chemical beyond intended targets (Rivas-Garcia et al., 2022; Dias et al., 2023).¹ At its peak, glyphosate was sprayed over an area roughly the size of Los Angeles (Camacho and Mejía, 2017). By 2015, mounting health concerns led to the suspension of the spraying program.

We use this policy setting to answer two questions. First, to what extent does wind drift from coca spraying damage legitimate, non-targeted crops? Second, does agricultural production recover after coca spraying was banned? By answering these questions, we quantify the spatial spillovers of place-based policy as well as the dynamics of agricultural recovery.

The main empirical challenge in quantifying spatial spillovers is that coca spraying is not random. Fumigation targeted coca-abundant areas, which systematically differ in terms of drug war-related violence, institutions, and poverty. Naive comparisons between municipalities more- and less-exposed to spraying would therefore confound off-target effects with these underlying differences. Causally estimating spillovers from spraying, hence, requires isolating variation in exposure that is orthogonal to the determinants of spraying.

¹While some genetically modified (GM) varieties are less sensitive to glyphosate, adoption of GM crops in Colombia during our study period was very low (< 1% of cultivated land) (Section 2.2)

We address this empirical challenge by computing a plausibly exogenous measure of wind-driven exposure to glyphosate spraying using a novel air particle transport model. We start by digitizing maps from the UN Office on Drugs and Crime (UNODC) that contain all coca plantations targeted by aerial glyphosate spraying between 2011 and 2015. We then deploy the dispersion model to simulate 3D wind pathways of tracer particles emanating from each sprayed area at a similar altitude as the actual aircraft. The fraction of tracer particles landing in a gridcell provides a continuous measure of a location's wind-driven exposure to herbicide drift. We run the model on a coca plantation-by-month basis for five years and aggregate to the municipality-year level. This yields a plausibly exogenous measure of municipalities' wind-driven exposure to aerial fumigation in a given year.

To estimate off-target damages from targeted spraying, we link our model-generated exposure measure to panel data on legitimate crop production from municipal agriculture surveys. This matched panel enables a two-way fixed effects (TWFE) design that gives rise to credible control groups for municipalities exposed to aerial spraying: other municipalities sharing similar characteristics, located equidistant from sprayed areas, but less exposed due to plausibly random differences in wind drifts.

Our main analysis yields three key findings. First, we document extensive damage to legitimate, non-targeted crops from wind-driven exposure to aerial coca spraying. Municipalities more exposed to glyphosate drift experience 13.2% lower harvests (hectares) compared to less exposed municipalities, with similar reductions in crop output (tons) and revenue (\$USD). We show a similar pattern of off-target crop damage in satellite-derived measures of vegetation quality. In a back-of-the-envelope calculation using these estimates, we document that Colombia's agricultural expansion was held back by roughly 12% due to off-target crop damages from aerial fumigation.

Second, we document heterogeneity in spillover damages. In particular, higher soil quality cushions spillovers from aerial spraying, whereas diversifying one's crop portfolio offers limited protection. This is consistent with glyphosate's broad-spectrum nature, which suggests that vulnerability to drift depends more on environmental conditions than on crop composition. In particular, clay-rich soils can bind glyphosate and reduce its bioavailability.

Third, we find little evidence of adaptation among farmers exposed to glyphosate drift in the medium run. If farmers adopted herbicide-resistant varieties or showed other adaptive behaviors, damages would diminish over time, producing a U-shape in lagged coefficients. Instead, cumulative lag estimates show that damages to off-target crops are unchanged three years later. We confirm this with municipality-level data on farm loans, using the idea that farmers may adapt by allocating credit to adaptive investments. Yet, we find no impact of exposure to aerial spraying on farm credit. Lastly, we show the lack of adaptation more directly using data on GMO crop adoption, although data is only available at the coarser

department level. We nevertheless find no correlation between glyphosate exposure and GMO crop shares. All these three pieces of evidence point to very limited adaptation.

The causal interpretation of our key results is subject to several identification concerns. One concern is that prevailing wind patterns might be foreseeable to government officials, raising the possibility that spray locations were selected partially based on expected downwind exposure. However, this is unlikely because the Colombian government's official position throughout the Plan Colombia era was that aerial glyphosate spraying caused no significant harm to people, agriculture, or the environment (DNE, 2003; Reyes, 2007; Republic of Colombia, 2010).² This position is also consistent with the international scientific view at the time, since the World Health Organization did not reclassify glyphosate as a probable carcinogen until 2015 (International Agency for Research on Cancer, 2015).³ Crucially, a government that believed spraying to be harmless had no reason to weigh expected downwind drift when choosing where to spray. Still, we complement these arguments with a placebo test that simulates particle paths from *non-sprayed* coca farms (Section 5.2). If authorities systematically avoided spraying coca whose drift would land on productive or politically connected farmland, then placebo exposure would inherit the same targeting bias and predict agricultural outcomes. Instead, the placebo has no impact on legitimate agriculture, suggesting that our estimates are driven by glyphosate drift rather than endogenous targeting or other characteristics of coca-growing regions. The placebo also rules out alternative airborne pollution sources at coca farms (such as biomass burning), since these would be present at both sprayed and non-sprayed farms.

A second related concern is that authorities may have strategically avoided spraying coca farms upwind of protected areas. However, controlling for the share of municipal land under protection leaves our estimates unchanged (Section 5.3). A third concern is that commercial fumigation of legal crops (e.g., bananas, rice) could affect both agriculture and our exposure measure if its drift coincided spatially with coca spraying.⁴ Using flight permits to construct a measure of commercial fumigation exposure, we control for this potential confounder, and our coefficient remains stable. A fourth concern is that differential agricultural trends across Colombian departments could threaten the design. Adding department-year fixed effects, which absorb such trends, leaves the coefficient essentially unchanged. We report additional robustness checks in Section 5.3.

Taken together, our findings from the Plan Colombia era provide new evidence that aerial

²In its 2010 Counter-Memorial to the International Court of Justice, Colombia stated that “there is no reliable evidence that any of the damage alleged has actually occurred, still less that it is attributable to aerial spraying; the evidence is to the contrary” (Republic of Colombia, 2010, ¶ 1.32).

³Moreover, spraying missions were operationally led by the National Anti-Narcotics Police (DIRAN; Section 2.2.2), whose mandate was coca eradication and operational security rather than agricultural protection.

⁴Standard agricultural fumigation differs from coca eradication in flight altitude and herbicide concentration, which reduces the risk of drift. See Section 2.2.2 for details.

coca eradication imposed widespread negative externalities on the legitimate agricultural sector. These damages are especially concerning given that spraying was largely ineffective at reducing coca cultivation ([Mejía, 2016](#)). Large spillover costs and limited policy effectiveness underscore clear inefficiencies in program design. In 2015, President Juan Manuel Santos terminated aerial spraying, citing a World Health Organization declaration about the harms of glyphosate exposure ([New York Times, 2015](#)). A decade later, it remains unclear whether undoing the policy reversed the damage already done.

In the second part of the paper, we extend the study period to 2018 and use a difference-in-differences design to test whether the off-target damages from aerial spraying persist after the policy ended. Treatment intensity is measured as cumulative pre-policy exposure to glyphosate, computed from the same dispersion model. The identification logic is as follows: if glyphosate drift caused only transitory damages, then municipalities that were highly exposed before 2015 should recover and converge back toward the trajectory of less-exposed municipalities after the ban. If damages were persistent, however, then previously-exposed municipalities should continue to lag behind their counterfactual trajectory. The coefficient on the interaction between pre-policy exposure and the post-2015 indicator thus captures the extent to which agricultural losses were reversed in the post-ban period.

Our difference-in-differences estimates show that the ban on aerial coca spraying triggered only a modest recovery of legitimate agriculture. Crop harvests in previously-exposed municipalities rose by about 2.3% post-ban, compared to the 13.2% decline incurred during the policy era. The magnitude of recovery remains small even four years after the ban, and we also find no corresponding recovery in crop revenue or satellite measures of vegetation health. These results indicate that collateral damage from spatially targeted agricultural enforcement is highly persistent and that stopping the policy does not necessarily restore the agricultural sector in non-targeted areas.

Given the persistence of agricultural damages, the question arises: what happened to damaged cropland that failed to recover? We investigate this by combining high-resolution land cover data with our difference-in-differences specification, using the municipality share under various land types as the outcome. We find that municipalities previously-exposed to glyphosate drift partly transition into grassland and shrubland after spraying stops.

The paper concludes with a timely policy analysis. In April 2025, under renewed U.S. pressure, the Colombian government announced plans to resume aerial fumigation ([Tactics Institute, 2025](#)) after its decade-long ban. Using our retrospective estimates, we simulate how much aggregate spillover crop damage can be avoided through smarter targeting the second time around. We find that \$USD 639 million of collateral damage on legitimate crops can be avoided compared to business-as-usual if spray campaigns only target the largest coca hotspots, all else equal. These savings are equivalent to roughly 3.7% of Colombia's

agricultural GDP, representing substantial gains from simple policy changes.

Our findings offer two main contributions to the literature. First, we credibly estimate *nationwide* externalities from a spatially targeted enforcement policy, providing new evidence on the indirect costs of place-based interventions. Recent studies document *nearby* spillovers from policies targeting crime (Blattman et al., 2021), overfishing (Englander, 2023), and deforestation (Assunção et al., 2023; Vieira et al., 2023), typically within pre-defined geographic buffers and over relatively short time horizons. We extend this work by deploying a chemical dispersion model that traces spillovers across the country rather than local buffer zones. We also move beyond the short-run and find new evidence that spillovers are persistent and outlast the policy itself.

Our second contribution extends the health economics literature by considering agricultural impacts. The health literature on pesticide externalities mainly documents adverse impacts on human capital (Agostini et al., 2020). Closest to our setting, Camacho and Mejía (2017) and Horta-Sáenz and Tami-Patiño (2024) study direct effects of aerial coca spraying within sprayed Colombian municipalities, and find worse health and educational outcomes, respectively. Related studies link pesticide exposure to poorer birth outcomes, higher childhood cancer incidence, and other long-run health costs in Ecuador, Brazil, and the United States (Calzada et al., 2023; Dias et al., 2023; Skidmore et al., 2023; Reynier and Rubin, 2025; Frank and Taylor, 2022; Frank, 2024).

Our paper provides the agricultural counterpart to this health literature. Existing work on herbicide drift is limited to U.S. and European contexts and mainly focuses on commercial applications (e.g., Roundup and dicamba) and highly localized spillovers. Larsen et al. (2024) estimate externalities from conventional herbicides onto nearby organic farms in California; Missirlian (2024) uses wind quadrants to identify dicamba drift and seed-adoption responses at the U.S. county level; Coinon (2025) combines upwind and upslope exposure for all pesticides at the parcel level in France; and Young et al. (2023) models yield losses and seed adoption from dicamba drift using a structural network-diffusion approach in two experiment fields in Missouri. Our contribution is in documenting a different setting where herbicides are deployed as a government enforcement tool rather than as a farm input. In our policy setting, targeting, timing, and dosage were determined by counter-narcotics objectives rather than agronomic considerations, applying several times the conventional per-hectare dose at far higher concentrations (Section 2.4). We thus provide the first evidence on the agricultural spillovers of a state-led, place-based enforcement policy, complementing the existing literature on the health consequences of aerial coca spraying.

These contributions are enabled by a methodological innovation that uses NOAA’s HYSPLIT atmospheric dispersion model to construct plausibly exogenous measures of wind-driven herbicide exposure across municipalities. With this advance, we join an emerging ap-

plied economics literature that uses dispersion models to trace particle drift in air-pollution applications (Heo et al., 2025; Morehouse and Rubin, 2026; Hernandez-Cortes and Meng, 2023; Wen et al., 2023; Abman et al., 2024); we are, to our knowledge, the first to apply this approach to measure herbicide drift onto non-targeted crops.

The next section provides the necessary background. Section 3 describes the data and dispersion model. Section 4 outlines the empirical strategy for evaluating spillovers from aerial spraying and Section 5 presents the results. Section 6 presents differences-in-differences estimates of the impact of banning aerial spraying. Section 7 simulates avoided damages from better targeting, and Section 8 concludes.

2 Background

This section provides background on Colombia’s agricultural sector, illicit coca cultivation, and the aerial spraying program aimed at eradicating coca.

2.1 Place-Based Policies in Agriculture

Agricultural place-based policies are defined by two features: they delineate a geographic boundary and condition policy action on that boundary. Governments use these policies to promote desired activities and to deter undesirable ones. On the promotion side, textbook examples include location-specific input subsidies, irrigation investments, and conservation payments. On the deterrence side, examples include land use restrictions near protected areas, pesticide restrictions near schools or water bodies, and enforcement campaigns against illicit agricultural activities—the subject of this paper.⁵

The defining feature of agricultural place-based policies is also their main challenge. While the policy targets a location, spillovers via prices, chemical inputs, and other factors can cross boundaries, generating unintended impacts on non-targeted areas (Hanson and Rohlin, 2013, 2021; Neumark and Simpson, 2015). These spillovers take two forms: within-sector spillovers displace the targeted activity itself, such as crime control shifting crime to neighboring areas (Blattman et al., 2021), or forest protections pushing deforestation into adjacent unprotected forests (Assunção et al., 2023). Cross-sector spillovers, by contrast, affect an entirely different activity altogether. Measuring cross-sector spillovers is important for assessing whether place-based policies deliver net social benefits.

⁵By contrast, non-place-based policies include supply-chain interventions, where authorities targeted precursor chemicals or distribution networks.

2.2 Agriculture in Colombia

2.2.1 Legitimate Agriculture

We focus on agricultural spillovers from aerial coca eradication because agriculture is a pillar of the Colombian economy: it occupies 40% of the country's land area (44 million ha), contributes 15% of GDP, and employs roughly 7.4% of the workforce, with smallholders responsible for 70% of domestic food production (OECD, 2015, 2023). Colombia's diverse landscapes feature a variety of crop systems potentially affected by glyphosate drift. In the tropical lowlands, banana, cotton, soybeans, rice, and sugarcane thrive. In mid-elevation temperate areas, farmers grow coffee, citrus fruits, papaya, flowers, and beans, many of which are critical for the country's agricultural exports. Cooler highlands support cereals (e.g., wheat, barley, oats), vegetables, tubers (e.g., potatoes, yucca), and fruit trees.

2.2.2 Illicit Agriculture: Coca Cultivation

Coca cultivation is the world's largest illicit agribusiness (Rincón-Ruiz and Kallis, 2013), with Colombia as the top supplier of coca leaves, the key input in cocaine production. In 2023, coca cultivation in Colombia reached 254,000 ha, roughly the size of Tokyo and the highest level in two decades (UNODC, 2024). Despite domestic and international eradication efforts, coca farming remains persistent, prompting militarized state-led intervention.

Multiple factors explain coca's resilience. Agronomically, the coca plant thrives in Colombia's fertile soils and favorable climate. Economically, the crop offers rural households a rare combination of short cultivation cycles (6 to 7 months), stable market demand, and on-farm sale to traffickers. In remote regions where infrastructure is sparse and state capacity limited, coca remains one of the few reliable sources of household income.

2.3 Plan Colombia: Aerial Coca Eradication

Successive Colombian administrations have experimented with a wide range of coca eradication policies, from voluntary crop substitution to forced manual removal. By the late-1990s, aerial herbicide spraying became the dominant strategy (Tokatlian, 1995, 1998). This approach was scaled up dramatically under Plan Colombia, a U.S.-backed counter-narcotics and defense program launched in 1999. During the program, roughly \$11.6 billion was directed toward eradicating coca through large-scale fumigation campaigns and strengthening the Colombian military and police. The National Anti-Narcotics Police (DIRAN) led fumigation missions, with logistical and security support from the military. Large aircraft such as the T-65 Thrush and Air Tractor AT-802 (Solomon et al., 2005) sprayed herbicides, while armed helicopter escorts deterred insurgent groups.

Glyphosate served as the primary input in aerial eradication campaigns. Chemically known as N-(phosphonomethyl) glycine, glyphosate is a broad-spectrum herbicide widely used in commercial agriculture (PAN, 2016; Benbrook, 2016). It disrupts key biological pathways necessary for plant growth (Giesy et al., 2000; Tzanetou and Karasali, 2020). Commercial formulations, such as Roundup, combine glyphosate salts with surfactants to improve leaf absorption. In Colombia, the sprayed mixture included Cosmo-Flux 411F (U.S. Department of State, 2002; Nivia, 2002; Hewitt et al., 2009), a surfactant blend designed to enhance penetration through waxy leaf cuticles (Solomon et al., 2005). While effective at increasing herbicidal potency, this formulation also raises the potential for unintended uptake by non-target crops and increases the likelihood of collateral damage.

2.4 Wind Drift and Off-Target Damages

Herbicide drift occurs through three mechanisms: i) primary (spray) drift, when droplets are carried by wind during application, (ii) vapor drift, resulting from post-application volatilization; and (iii) particle-bound drift, arising from post-application environmental processes such as wind erosion (van den Berg et al., 1999; Bish et al., 2020). Although glyphosate is non-volatile and therefore not subject to vapor drift (Franz et al., 1997), it remains susceptible to spray drift during application and may also become airborne when attached to soil or dust particles (Tzanetou and Karasali, 2020).

Several factors influence the extent of glyphosate drift: droplet size, nozzle height, wind speed, atmospheric stability, and soil moisture (Jordan et al., 2009; Fritz et al., 2010; Bish et al., 2020). Fine droplets (< 200 microns) are particularly prone to being carried downwind, while dry soils exacerbate dispersion by promoting dust-bound glyphosate movement (Mef-taul et al., 2021). Under intense precipitation events, rainfall can remove up to 97% of airborne glyphosate, redistributing it onto non-target surfaces (Chang et al., 2011). In aerial applications, operational parameters also matter: higher aircraft speeds generate smaller droplets (Hewitt et al., 2009), and doubling nozzle height can triple downwind deposition (Butts et al., 2022). Field evidence indicates that low-altitude spraying produces 3 to 8 times more drift than ground-based methods (Butts et al., 2022).⁶

Colombian fumigation operations exhibited conditions highly conducive to drift. Aircraft often flew approximately 30-50 meters above ground (Solomon et al., 2005; DIRAN, 2020; U.S. Department of State, 2003), far higher than the 2-3 meter spraying altitudes in conventional agricultural spraying. Although official protocols restricted spraying under adverse conditions⁷ and required coarse droplets, compliance proved difficult in practice

⁶Despite mitigation efforts such as low-altitude flight and coarse droplets (>300 microns), glyphosate residues have been detected up to 800 meters from target areas (PAN, 2016; Ravier et al., 2019).

⁷Protocols prohibit spraying after rainfall, wind speeds > 2 m/s, and humidity < 75% (Solomon et al., 2005)

Table 1: Comparison of Glyphosate Application Rates

Source	Glyphosate applied (kg/ha)
<i>Commercial agriculture (average)</i>	
INL (2002)	< 0.75
Benbrook (2016)	≈ 1.0
<i>Aerial coca eradication</i>	
Vargas et al. (2003)	3.73
Solomon et al. (2005)	4.99
Marshall et al. (2009)	3.70
MinAmbiente and MinSalud (2014)	4.99

Note: Figures are total glyphosate active ingredient applied per hectare. Commercial-agriculture rates are U.S. averages (less than 0.75 lb a.i./acre, [U.S. Department of State, 2002](#); [Benbrook, 2016](#)). Coca-eradication rates are computed from reported spray volumes and mixture concentrations. Aerial coca eradication thus applied roughly four to six times the conventional per-hectare rate.

([Nivia, 2002](#)). Moreover, despite GPS navigation and real-time weather monitoring, pilots retained final discretion over mission parameters ([Solomon et al., 2005](#)).

Glyphosate application dosage and concentration also far exceeded conventional agricultural guidelines.⁸ Colombian technical assessments indicate that aerial fumigation applied roughly 3.7-5.0 kg of glyphosate active ingredient per hectare (Table 1), about four to six times the average application rate in commercial agriculture ([INL, 2002](#); [Benbrook, 2016](#)). The sprayed mixture was also far more concentrated than in conventional use—roughly 26 times the glyphosate concentration recommended for normal weed control—and the Cosmo-Flux 411F surfactant further amplified biological efficacy by up to a factor of four ([Nivia, 2002](#)). The combination of substantially higher application rates, a more concentrated and surfactant-enhanced mixture, and elevated flight altitudes make it critical to understand the extent to which airborne glyphosate drifted beyond its intended target and caused off-target damage in receiving regions.

2.5 The End of Aerial Spraying

Despite intensive fumigation campaigns under Plan Colombia, coca cultivation remains persistent, casting doubt on the long-term effectiveness of chemical eradication policies. While aerial spraying was operationally effective in delivering herbicide to targeted coca plots, mounting complaints by rural households highlighted substantial risks, including herbicide exposure, excessive dosages, and significant harm to rural communities ([Defensoría](#)

⁸[Marshall et al. \(2009\)](#) shows that low glyphosate doses provide poor control of coca.

del Pueblo, 2014). These issues fueled public opposition and policy debate, prompting skepticism about the strategy’s viability as a counter-narcotics tool.

Starting in the mid-2000s, accumulating evidence began linking glyphosate spraying to a range of environmental and health impacts. Research documents contamination of drinking water (Myers et al., 2016), damages to soil microbiota (Dewar et al., 2000; Sanderson et al., 1999), disruption of pollinator behaviors (Delkash-Roudsari et al., 2020), and impaired plant nutrient uptake and seed germination (Blackburn and Boutin, 2003).

These growing concerns culminated in 2015, when Colombia officially banned aerial coca spraying. The decision followed the World Health Organization’s classification of glyphosate as a probable human carcinogen (International Agency for Research on Cancer, 2015). Whether the ban reversed the damages accrued over years of intensive fumigation remains an empirical question. We answer this question in Section 6 and show that damages persist several years after the policy ends.

3 Data

We assemble several novel datasets to quantify spillovers from aerial coca spraying, including locations of illegal plantations, digitized maps of sprayed fields, and municipal agricultural surveys. Wind drift from aerial spraying is measured via an atmospheric dispersion model. This section provides an overview of the data and dispersion model.

3.1 Data Sources

3.1.1 Illicit Coca and Aerial Spraying

We obtained 1km resolution maps of illicit coca plantations between 2011-2015 from the UNODC Integrated Illicit Crops Monitoring System (SIMCI in Spanish). UNODC processes and verifies raw imagery to distinguish coca from other crops. The final dataset measures hectares of coca in a gridcell. Figure 1A (green) maps coca in the baseline year. About 20% of municipalities grow coca, mainly in the North, West, and Central regions. The average municipality cultivates about 3,300 ha per year of coca (Table A1).

To measure aerial spraying, we digitized SIMCI’s annual Coca Cultivation Survey maps of sprayed areas from 2011 until 2015 (UNODC and Government of Colombia, 2011-2015).⁹ Shapefiles demarcate polygons sprayed during the year. Since our concern is wind drift

⁹We use 2011 as the starting year because it is a post-adjustment SIMCI census year. The SIMCI’s official 2011 coca census incorporates the small-plot adjustment used to maintain comparability in the historical series after coca plots became smaller on average. In 2015 the Colombian government suspended the aerial fumigation program (see Section 6).

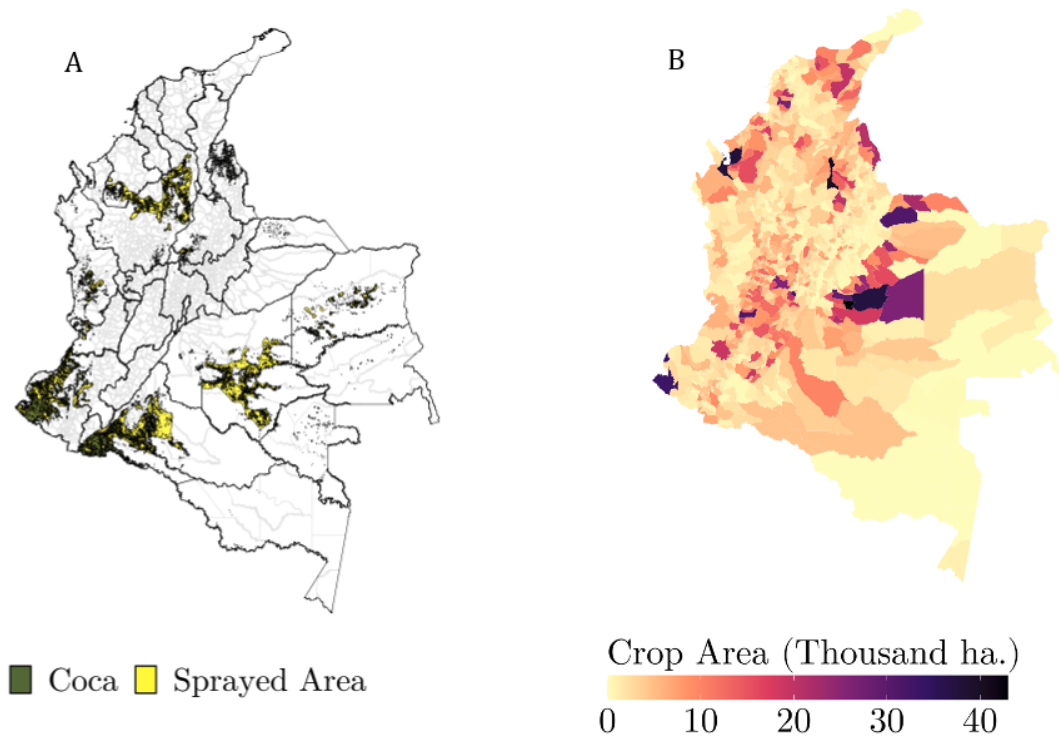


Figure 1: Coca Plantations, Aerial Spraying, and Legitimate Agriculture (2011)

Note: Data are from 2011. In Panel A (left), green areas are coca and yellow are sprayed areas—all sprayed areas had coca planted. Panel B (right) shows harvested area of legitimate crops across municipalities.

from aerial spraying, these polygons serve as the point source for our dispersion model (Section 3.2.1). Figure 1A (yellow) shows sprayed areas in the baseline year: roughly 5,600 ha per year were sprayed in the typical municipality (Table A1). It can also be seen that not all coca fields are sprayed, implying strategic targeting. While the exact selection criteria employed by the National Anti-Narcotics Police are unknown, we address this “place-based endogeneity” as best we can with a combination of controls, balance tests, and falsification exercises to ensure that more- and less-exposed municipalities are comparable (Section 4.2).

3.1.2 Legitimate Crop Production

To quantify spillovers from targeted herbicide spraying onto non-targeted crops, we use detailed municipal-level agricultural data from the Ministry of Agriculture’s Evaluaciones Agropecuarias Municipales (EVA) survey from 2011 to 2018¹⁰. This dataset reports planted area, harvested area, and output for 223 crops at the crop-municipality-year level, making it

¹⁰Until 2018, EVAs were collected and consolidated by the Ministry of Agriculture and Rural Development. In 2019, responsibility for the collection, validation, and processing of this statistical operation was transferred to the Rural Agricultural Planning Unit, alongside the introduction of a centralized web-based municipal reporting platform. As a result, post-2018 EVA data reflect a different institutional and reporting framework.

the most granular and comprehensive sub-national agricultural dataset for Colombia. Figure 1B maps harvested crop area: the typical municipality produces 44,000 tons of crops on 4,000 ha of land (Table A1). The figure also shows that most coca grows outside of agricultural hubs, consistent with coca’s illicit status, which pushes production to remote, hard-to-monitor areas. Yet, as we show in Section 3.2.1, coca hubs are connected to agricultural hubs via wind drifts, exposing productive farmland to glyphosate through aerial coca spraying.

Besides EVA, we also use administrative data on the share of harvested area planted with genetically modified maize and cotton, obtained from the Instituto Colombiano Agropecuario (ICA) through a Freedom of Information request. This data is at the department-year level, and we use it only in Section 5.5 to test for substitution into glyphosate-tolerant varieties.

We calculate the economic value of crops in our sample using FAOSTAT, which records crop prices received by farmers in \$USD/ton at the national level. First, we identify 51 crops grown in Colombia for which FAOSTAT price data are available. The crop prices cover crops accounting for about 97% of national agricultural production by weight. For each category-year, we compute annual revenue by multiplying its 2011 price (to express values in constant USD) by total production in tons. Lastly, we aggregate crop revenues at the municipal level to obtain total crop revenue per municipality per year.

To validate our estimates from administrative data, we compute satellite measures of crop conditions. First, we use the Normalized Difference Vegetation Index (NDVI), a widely used measure of vegetation health, obtained from MODIS at 1km resolution (Didan et al., 2015).¹¹ Second, we use remotely sensed land classification maps at 300m resolution from the European Space Agency (ESA). We use the cropland layer, which indicates if a pixel has rainfed, irrigated, or mosaic crops (small-scale agriculture interspersed with natural vegetation). To obtain a municipality-level measure, we extract means over cells within a municipality, weighted by cell overlap fraction.

3.1.3 Covariates

We include two sets of covariates in all our regressions. The first set accounts for the endogenous spraying decision and includes coca area and distance to the nearest sprayed area. Coca area accounts for the fact that places with more coca are more likely to be sprayed. Distance from the centroid of each municipality to the nearest sprayed area controls for broader spatiotemporal differences between places close and far from eradication zones, including political violence, remoteness, and other factors that may co-determine the spraying decision and agricultural outcomes. We diagnose any remaining selection bias with a variety of balance tests and falsification exercises in Section 4.2.

¹¹Since NDVI also captures forests, we control for forest cover in all regressions with NDVI using forest cover data from the Vegetation Continuous Fields (VCF) gridded data product (Townshend et al., 2017).

The second set of covariates includes temperature and rainfall, which partially guide wind drifts and can affect agriculture directly (Yuan et al., 2023). Annual temperature (°C) and rainfall (mm) are from the ERA5 product at 0.125° resolution (Hoffmann et al., 2019). In a robustness test (Section 5.3), we augment this set with additional meteorological controls, including energy flux, planetary boundary layer height, humidity, and velocity, and find no changes in our estimates. For all gridded data, we extract means over cells within a municipality, weighted by cell overlap fraction.

3.2 Measuring Wind-Driven Exposure to Aerial Herbicide Spraying

Our task is to measure the exposure of a municipality to aerial spraying. Our starting point is the fact that aerial pesticide spraying is prone to particle drift (Boonupara et al., 2023; Cederlund, 2017; Gandhi et al., 2021). To measure the movement of airborne pesticides, we use an atmospheric dispersion model that simulates the most likely drift pathways from a source location. We summarize the basics of the model below and leave technical details on parameterization, deposition, and diffusion in Appendix B.1.

3.2.1 The HYSPLIT Dispersion Model

We use the Hybrid Single-Particle Lagrangian Integrated Trajectory (HYSPLIT) model developed by NOAA (Draxler and Hess, 1998; Draxler et al., 2020) to measure municipalities' exposure to aerial spraying via wind drift. The model simulates three-dimensional movement of tracer particles released from a point source—in our case, the location sprayed. Trajectories are primarily driven by wind, but also incorporate other meteorological data.¹² As described below, we measure a municipality's exposure to aerial spraying by the share of tracer particles originating from sprayed areas landing within its borders.

The main advantage of measuring exposure in this way is that we can precisely measure spillovers from aerial spraying. Since particle paths from sprayed locations can pass over and deposit in municipalities that were not sprayed (see Appendix B.1 for details), these non-sprayed municipalities are exposed to glyphosate entirely through wind-driven spillovers. Whether spillovers lead to crop damage is the central question of this paper.

There are several related advantages of using a dispersion model. First, the shape, size, and diffusion rate of dispersion plumes are entirely model-generated. In contrast, traditional wind direction vectors require assuming an arbitrary length and plume, often a 45° cone around the vector. Second, our measure can be cumulated to any spatial or temporal level without loss of information, which becomes important when defining treatment

¹²HYSPLIT also uses temperature, planetary boundary height, surface pressure, velocity, surface roughness, heat flux, short wave flux, humidity, convective energy, and precipitation (Draxler and Hess, 1998).

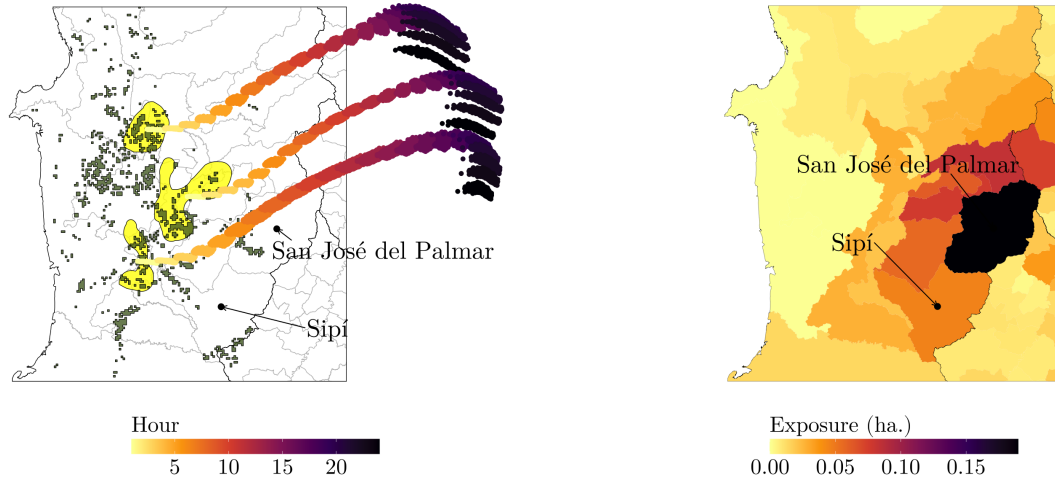


Figure 2: HYSPLIT Output (January 2011)

Note: Panel A shows HYSPLIT particle trajectories on January 1st, 2011 from three example sprayed areas (yellow). Colors denote where particles end up in each hour. Panel B shows municipality exposure (E_{mt} in Equation 2) to coca eradication in 2011.

and control groups in our difference-in-differences design (Section 6.1). Conventional wind vectors capture seasonal variation in wind direction, which loses explanatory power when averaged over multiple years. Lastly, since our exposure measure is continuous, we can group similar values into “plumes” and correct for spatial error correlation by clustering at the plume level. This enables more precise estimation of standard errors.

3.2.2 Measuring Exposure to Aerial Spraying

Step 1: Dispersion The first step for computing exposure to aerial spraying is to measure wind drift from each sprayed area. We generate a 0.1° grid over Colombia and set the centroid of each sprayed area (Figure 1A) as the point source.¹³ From each point source, we set HYSPLIT to emit a default number of tracer particles from 50m height for 24 hours, once a month, from 2011-2015.¹⁴ Appendix B.1 provides more detail on these parameterizations. We make these modeling choices in the absence of data on the exact days that spraying took place. Running the model once a month enables us to calculate average statistics of wind drift over the year. While less precise than if we knew exact spray dates, using the prevailing wind direction during the year is common in the literature when working with annual data (Hernandez-Cortes and Meng, 2023; Wen et al., 2023; Abman et al., 2024; Heo et al., 2025; Morehouse and Rubin, 2026).

Figure 2A shows hourly tracer particle locations from an example spray in 2011. Note

¹³For big spray areas (>95th pctl), we randomly sample 5 points on the polygon as the point sources.

¹⁴50m is the typical height at which aircraft drop glyphosate (U.S. Department of State, 2003)

that these particles do not represent glyphosate per se—they are tracers that draw the most likely path that any particle from the point source would take given HYSPLIT meteorology assumptions. We implement placebo tests in Section 5.2 to rule out other emission sources in the sprayed area, strengthening the case that particle paths trace glyphosate drift. After each model run, we measure exposure of a gridcell as the downwind count of tracer particles, $ParticleCount_{ijt}$, landing in receptor gridcell i , originating from spray location j at time t . Since the model is run monthly for each sprayed area, j indexes model runs rather than unique spray locations. Let $\mathcal{J}_t$ denote the set of all twelve model runs in year t .

Step 2: Normalization The second step is to normalize $ParticleCount_{ijt}$ to one and then sum over model runs j in each year, adjusting for differences in sprayed area, $SprayArea_{jt}$:

$$CellExposure_{it} = \sum_{j \in \mathcal{J}_t} \left(\frac{ParticleCount_{ijt}}{\sum_i ParticleCount_{ijt}} \cdot SprayArea_{jt} \right) \quad (1)$$

Since we sum over all sprayed areas j in a year t , $CellExposure_{it}$ measures the *total modeled exposure* of gridcell i to aerial spraying in that year. This measure also accounts for the fact that larger spray zones pose greater potential for exposure. $CellExposure_{it}$ can be interpreted in ha, since $SprayArea_{jt}$ is effectively distributed across gridcells according to cell i 's exposure intensity, which ranges from zero to one. For example, if the area sprayed is 100 ha, and the exposure of a far-away gridcell i is 0.01, then it is *as if* 1 ha of coca was sprayed in gridcell i , even though it may have no coca. This is how we capture spillovers.

While $CellExposure_{it}$ does not explicitly incorporate particle deposition, exposure and deposition are related. Appendix B.1 explains the connection in more detail.

Step 3: Aggregation The third step aggregates at the municipality-year level. Municipality m 's exposure to spraying in year t is the sum of modeled exposure across gridcells $i \in m$:

$$E_{mt} := \sum_{i \in m} CellExposure_{it} \quad (2)$$

E_{mt} is the main explanatory variable in the first part of the empirical analysis, and is visualized in Figure 2B. The underlying variation in E_{mt} arises from three sources. First, *spatial variation* arises from physics formulae for gas movement and diffusion as modeled by HYSPLIT. A second source of *spatial variation* comes from the size of the area sprayed. Holding meteorology constant, the spraying of larger coca plantations generates greater exposure due to the area-weighting in Equation 1. Third, *temporal variation* arises from embedded formulae that describe the dynamics of particle transport. Importantly, both spatial and temporal variation depend on the location sprayed and can change when new plantations

are targeted or previous spray campaigns are halted. This means that exposure is only plausibly exogenous *conditional on spraying*, since the choice to spray a location is strategic. We provide tests for this identification assumption in Section 4.2.

Again, E_{mt} does not measure exposure to glyphosate per se, but rather wind drift into municipality m from locations where coca was sprayed with glyphosate. Since these sprayed coca fields are often in remote places with little other economic activity, it is unlikely that E_{mt} captures exposure to any other airborne pollutant besides glyphosate. We confirm this in Section 5.2 with placebo tests, and by showing no impacts of E_{mt} on SO_2 , NO_2 , or $\text{PM}_{2.5}$.

4 Empirical Strategy

We use a TWFE design to compare agricultural outcomes within municipalities at varying levels of wind-driven exposure to coca spraying, controlling as best we can for strategic targeting of coca farms. This design exploits rich cross-sectional and temporal variation, giving rise to credible control groups for municipalities exposed to spraying: other municipalities with similar geographic characteristics, located equidistant from sprayed areas, but less exposed to them due to plausibly random differences in wind drifts.

Figure 2 provides visual intuition for our empirical strategy. Consider the experience of San José del Palmar and Sipí, two remote municipalities near the West coast. Both grow coca (Panel A, green), yet neither experienced aerial eradication in 2011 (Panel A, yellow). Moreover, they are both equidistant to the nearest sprayed area, meaning that they likely share similar determinants of spraying. Despite similar exposure *potential*, proximity, and geography, the two municipalities face drastically different exposure to the nearby spray campaign. San José lies directly in the exposure path of the southernmost sprayed area, whereas Sipí is completely unexposed. These differences can be seen even when aggregating exposure at the annual level (Figure 2B). Because differences in exposure arise from plausibly random wind patterns *conditional on the decision to spray*, we can use this variation to estimate the causal impact of downwind exposure to aerial eradication.

4.1 Main Estimating Equation

Our estimation strategy generalizes the visual intuition described above. We quantify spillovers from targeted coca spraying on off-target crops with the following equation:

$$\text{Log } Y_{mt} = \beta \cdot E_{mt} + \mathbf{X}'_{mt}\Omega + \alpha_m + \gamma_t + \epsilon_{mt} \quad (3)$$

where m indexes municipalities and t indexes years between 2011-2015. Y_{mt} denotes the

harvested area in municipality m at time t . We also estimate versions where Y_{mt} measures output, revenue, and satellite measures of crop health. E_{mt} measures m 's exposure to aerial spraying in year t (Equation 1). $\mathbf{X}'_{mt}$ is a vector of covariates including temperature, rainfall, and distance from m to the nearest sprayed area. The distance term accounts for differences between places close and far from sprayed farms that may co-determine the spraying decision and Y_{mt} . We address additional selection concerns in Section 4.2. All specifications include municipality and year fixed effects, α_m and γ_t , respectively.

The coefficient of interest is β , which captures the percentage change in agricultural outcomes from an additional hectare of wind-driven glyphosate exposure. $\beta < 0$ indicates negative spillovers, consistent with targeted aerial coca spraying harming off-target crops via wind-driven glyphosate drift. We cluster ϵ_{mt} at the plume level since gridded values of $CellExposure_{it}$ may be spatially correlated through unobserved formulae used by HYSPLIT that govern particle movement. Plumes are demarcated using a k-means clustering algorithm that groups cells with similar exposure values. Municipalities spanned by the same plume are assigned a common cluster ID and those spanned by multiple plumes are assigned the cluster with the largest overlap. We demarcate 50 plumes in the main analysis and test robustness to alternative numbers of clusters in Section 5.3. We also document robustness of our results to the computation of Conley standard errors.

β is estimated via OLS in a TWFE setup with continuous treatment (Callaway et al., 2024). This estimator suits our context because some municipalities receive continuous “doses” of exposure in each period, whereas others are untreated. β thus represents a dose-response to exposure at the municipality level and is best interpreted as the slope of the dose-response curve. We implement sensitivity tests following Callaway et al. (2024) in Section 5.3.

4.2 Identification

Identification of β relies on the assumption that exposure is orthogonal to unobserved determinants of crop output, once we control for municipality and year fixed effects. The main threat is that the decision to spray coca is endogenous, a concern exacerbated by the fact that the targeting criteria used by the National Anti-Narcotics Police are publicly unavailable. We partly address this by controlling for distance to the nearest sprayed area, which accounts for broad differences between places close and far from sprayed areas, including violence, remoteness, and other factors co-determining the spraying decision and crop outcomes.

However, the distance control alone does not eliminate selection bias. For example, if authorities avoid spraying coca upwind of agricultural regions, then variation in downwind exposure may partly capture differences in underlying agricultural trajectories, violating the assumption that $E[\epsilon_{mt}|E_{mt}] = 0$. To evaluate this concern, Table A2 presents a test

for selection bias by regressing exposure on lagged municipality characteristics, including agricultural area, weather, and political violence. While lagged coca predicts exposure—which is unsurprising since spraying targets coca-growing regions—we find no evidence that lagged agricultural outcomes, weather, or political violence systematically predict exposure. Results are similar when we apply a 30km mask around sprayed areas (column 2). This suggests that municipalities are not differentially exposed to glyphosate because they were already experiencing agricultural decline or differential shocks.

In Section 5.2, we provide an extra check against endogenous targeting by constructing a placebo exposure measure based on *non-sprayed* coca plantations. If downwind exposure is correlated with unobserved factors that influence the spraying decision, then placebo exposure should also predict agricultural outcomes. We find no such relationship, lending further support for the identification assumption that exposure is orthogonal to ϵ_{mt} .

Another threat to identification is that differences in weather between more- and less-exposed municipalities may affect particle trajectories as well as crop yields directly. We thus include temperature and rainfall in $\mathbf{X}'_{mt}$ to avoid potential omitted variable bias. In a robustness check, we include a variety of additional meteorological variables. Lastly, since exposure is based on wind paths, it may inadvertently capture pollution from other nearby sources such as energy, vehicles, or biomass burning. To rule out these alternative pathways, we show null impacts of exposure on PM_{2.5}, SO₂, and NO₂ (Section 5.3).

5 Results: The Plan Colombia Era

This section quantifies off-target impacts of targeted aerial coca spraying. We find that aerial spraying damages off-target crops via glyphosate drift, resulting in large revenue losses for farmers. We first present our main estimates and then investigate whether crop health rebounded after aerial spraying was banned in Section 6.

5.1 Main Results

Table 2 presents estimates of Equation 3. The main result is that $\beta < 0$: wind-driven exposure to targeted aerial coca spraying causes substantial damage to off-target crops. The point estimate in our preferred specification (column 2), which controls for distance to the nearest sprayed area, implies that an additional unit of exposure to aerial spraying during the year causes a 13.2% reduction in harvested crop area.

To put this estimate in perspective, consider the fact that crop harvests in the average municipality increased by 273 ha between 2011-2015, implying a pattern of agricultural growth across Colombia. At the same time, the average municipality was exposed

Table 2: Agricultural Spillovers from Coca Eradication

	(1) Area	(2) Area	(3) Output	(4) Revenue	(5) NDVI
Exposure (ha.)	-0.131** (0.055)	-0.132** (0.057)	-0.125** (0.050)	-0.141*** (0.040)	-0.014** (0.005)
Coca Controls	No	Yes	Yes	Yes	Yes
Climate Controls	Yes	Yes	Yes	Yes	Yes
Forest Cover	No	No	No	No	Yes
Municipality FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	5610	5610	5610	5610	5610

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. Outcomes are in logs. “Coca Controls” include distance to nearest sprayed area and hectares of coca. “Climate Controls” include temperature and rainfall. “Exposure” to aerial spraying is measured in hectare-equivalents. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

to 0.067 ha of airborne glyphosate via wind drift, translating into a reduction of harvested area by $e^{(0.067 \times -0.132)} - 1 = 0.88\%$ relative to the counterfactual without glyphosate drift. Given that the average municipality harvested roughly 3,703 ha of crops annually, this corresponds to about 33 ha of lost harvest area. Therefore, agricultural expansion experienced a $33/273 \approx 12\%$ setback due to spillovers from coca eradication between 2011 and 2015.

Columns 3 and 4 explore alternative outcomes: log output (tons) and log revenue (\$USD), respectively. Revenue is a weighted average of individual crop output and corresponding national prices (Section 3.1.3). Exposure to aerial spraying reduces output and revenue by magnitudes similar to the decline in harvested area. Column 5 validates our estimates with satellite-derived NDVI, a measure of vegetation health based on light absorbed by plants. Since NDVI measures both crops and forests, we control for forest cover in this specification to isolate effects on crop health. We continue to observe negative spillovers of aerial spraying on crop health, reinforcing our main estimates and providing reassurance that they are not driven by biases in administrative data collection.

5.2 Placebo Tests

Recall that exposure, E_{mt} , is based on *wind drifts* from areas sprayed with glyphosate, not actual glyphosate concentrations. This poses two identification concerns: first, the spraying decision is endogenous and, second, wind drifts may carry other airborne pollutants besides glyphosate. We present falsification tests below to help address both concerns.

We first simulate wind drifts from 200 randomly selected coca farms that were *not sprayed*

with glyphosate¹⁵, following the same modeling approach in Section 3.2. This generates a placebo measure of municipality m 's exposure to non-sprayed coca farms in time t . This placebo serves two purposes. First, it tests for endogenous targeting. If authorities avoid spraying certain coca farms based on unobserved characteristics of downwind municipalities, then placebo exposure should inherit this targeting bias and predict downwind agricultural outcomes even though glyphosate was never applied. Second, it tests whether crop damages (Table 2) are driven by airborne pollutants other than glyphosate. If pollution sources near coca farms (e.g., biomass burning) drive our results, then placebo exposure should still predict crop damages even though coca was not sprayed. Note that the placebo need not be a perfect counterfactual for sprayed coca farms; its purpose is simply to test whether being downwind of coca, absent spraying, is associated with crop damages.

Table A3 presents estimates where we directly control for placebo exposure in Equation 3. Both exposure measures are reported in standard deviations to account for differences in measurement. The placebo coefficient is near-zero and statistically insignificant for all outcomes, whereas the main coefficient of interest remains negative and significant. Combined with the balance tests in Section 4.2, this exercise adds confidence that our main estimates capture glyphosate drift rather than endogenous targeting or emissions from other sources.

As an additional safeguard, we show that our exposure measure does not entangle other pollutants by directly estimating impacts on SO₂ (associated with energy generation), NO₂ (associated with vehicle emissions), and PM_{2.5} (associated with biomass burning). Anecdotally, this endogeneity concern is minimal since coca plantations are set up far from human activity. Econometrically, if exposure is orthogonal to concentrations of these three pollutants, we can rule out alternative pollution pathways and become more confident that glyphosate drift is the key mechanism driving spillover crop damage.

Columns 1-3 of Table A4 show a near-zero and statistically insignificant impact of wind-driven exposure to aerial spraying on ambient PM_{2.5}, SO₂, and NO₂ concentrations¹⁶. Since the dispersion model measures wind paths from the exact locations sprayed, and there is no relationship between exposure to these locations and ambient pollution, it is hard to imagine any other reason driving crop damage in Table 2 besides glyphosate drift.

5.3 Robustness Checks

Table A5 tests robustness to alternative specifications (column 1-5), data sources (columns 6), and sample restrictions (columns 7-8). We begin by addressing concerns that spraying

¹⁵We sample 200 placebo farms to balance computational time and statistical precision. HYSPLIT is run on each farm using the steps in Section 3.2, with coca area as weights (Equation 1) instead of spray area.

¹⁶Gridded PM_{2.5} is obtained from Van Donkelaar et al. (2016). Gridded SO₂ and NO₂ are obtained from the MERRA-2 and HAQSTAT reanalysis products, respectively.

decisions are correlated with omitted characteristics of municipalities. One possibility is that authorities strategically avoid spraying coca farms upwind from protected areas. While the placebo test already provides evidence against this concern (Section 5.2), column 1 directly controls for the share of municipal land under protection using data from the World Database on Protected Areas. Estimates are highly stable, suggesting minimal bias from selection on protected area status.

We next consider alternative sources of herbicide exposure that may confound our estimates. Legitimate crop fumigation for rice, bananas, and other crops could affect agriculture while correlating with our exposure measure. Column 2 controls for the number of non-coca fumigation flights potentially crossing each municipality, constructed from AeroCivil permit data on flight dates and airport coordinates.¹⁷ The coefficient is virtually unchanged, consistent with legal agricultural fumigation usually being conducted at substantially lower altitudes and with smaller doses than aerial coca eradication (Section 2.4).

A related concern is that meteorological conditions determining particle paths may also affect crops directly. Column 3 augments our baseline controls with energy flux, planetary boundary layer height, wind velocity, and humidity.¹⁸ The coefficient remains very similar, suggesting that weather conditions are not confounding our estimates.

Column 4 adds department-year fixed effects to account for differential agricultural trends across departments. The coefficient remains within the confidence interval of the baseline estimate, which is unsurprising since exposure is plausibly orthogonal to department characteristics. Column 5 adds linear municipality time trends to control for unobserved factors that vary at a constant rate over time. The coefficient is very similar.

Column 6 validates our findings using an alternative satellite-derived measure of cropland extent to complement NDVI. Cropland percent is measured from ESA maps (Section 3.1.2) and includes irrigated crops, rainfed crops, or small-scale cropland interspersed with natural vegetation. The coefficient remains negative and statistically significant.

Finally, columns 7 and 8 address the concern that coca-abundant areas are both more likely to be sprayed and more likely to be exposed (Table A2). Column 7 applies a 30km mask around the sprayed area centroid¹⁹. This removes places in the vicinity of the sprayed area which may feature unobserved characteristics that co-determine spraying, exposure, and agriculture, thereby helping isolate wind-driven exposure. Column 8 adopts a more extreme approach by excluding all sprayed municipalities from the sample, which leaves the coefficient to be identified solely off of wind drift rather than endogenous coca eradication. We continue to find strong evidence of negative spillovers in both cases.

¹⁷We assume that legal fumigation is carried out on similar aircraft as coca fumigation. Through discussions with Anti-Narcotics officers, Reyes (2014) find that fumigation planes have a range of 80 miles.

¹⁸These data are from the MERRA-2 reanalysis product M2T1NXFLX files.

¹⁹The mask deletes gridcells within 30km of the point source during calculation of E_{mt} (Equation 1)

Table 3: Heterogeneity by Soil Quality

	(1) Area	(2) Output	(3) Revenue	(4) NDVI
Exposure (ha.)	-0.964** (0.459)	-1.030** (0.421)	-0.914*** (0.305)	0.006 (0.017)
Exposure (ha.) $\times$ Soil Quality	0.027* (0.015)	0.029** (0.014)	0.025** (0.010)	-0.001 (0.001)
Coca + Climate Controls	Yes	Yes	Yes	Yes
Forest Cover	No	No	No	Yes
Municipality FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	5610	5610	5610	5610

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. Outcomes are in logs. “Coca Controls” include distance to nearest sprayed area and hectares of coca. “Climate Controls” include temperature and rainfall. “Exposure” to aerial spraying measured in hectare-equivalents. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

Appendix B.2 provides additional robustness tests of the TWFE estimator following Callaway et al. (2024). We also report estimates from alternative methods of inference. The precision of our estimates is robust to alternative methods of accounting for spatial correlation.

5.4 Treatment Heterogeneity

We expect limited heterogeneity across crop types because glyphosate is a broad-spectrum herbicide that disrupts a biological pathway common to most plant species (Benbrook, 2016; Giesy et al., 2000; Tzanetou and Karasali, 2020). The main exception is glyphosate-tolerant GMO varieties, which were rarely grown in Colombia during our study period (Section 2.2). Any treatment heterogeneity should thus arise from environmental factors that affect how much glyphosate reaches plant tissues. A natural candidate is soil chemistry, because clay-rich soils bind glyphosate and reduce its bioavailability (Wang et al., 2019; Tzanetou and Karasali, 2020), potentially cushioning the damage from a given level of airborne exposure.

We test for treatment heterogeneity by soil quality with the equation below. We also provide diagnostic checks of the broad-spectrum nature of glyphosate by showing that exposure affects all crop types uniformly. We estimate:

$$\text{Log } Y_{mt} = \beta_1 \cdot E_{mt} + \beta_2 \cdot (E_{mt} \cdot S_m) + \mathbf{X}'_{mt} \Omega + \alpha_m + \gamma_t + \epsilon_{mt} \quad (4)$$

where m and t are indices for municipality and year, respectively. The term S_m measures

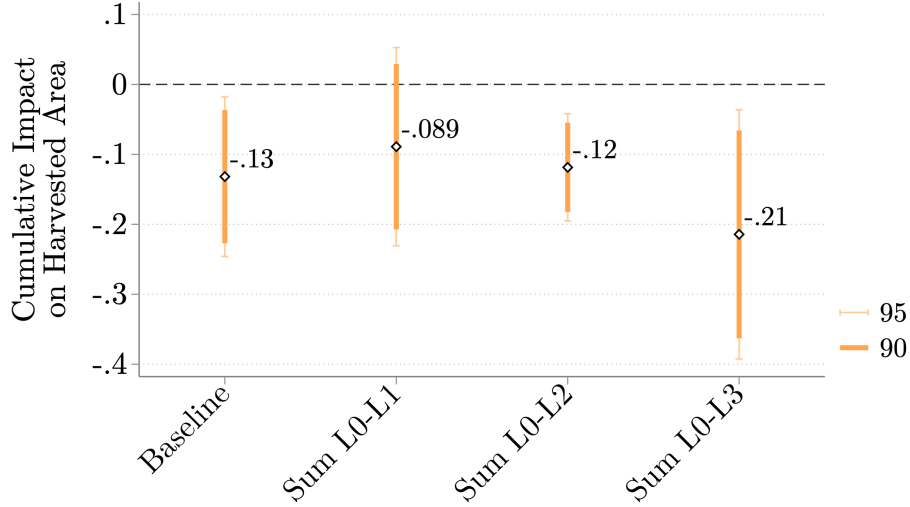


Figure 3: Dynamic Estimates

Note: The outcome is log harvested area. “Baseline” repeats the main result. “Sum L0-L1” adds the first lag of exposure to the main specification and reports the sum of coefficients on the first lag and baseline effect. “Sum L0-L2” sums up to the second lag, and so on. Bars are confidence intervals. All regressions control for distance to nearest sprayed area, hectares of coca, temperature, rainfall and municipality and year fixed effects. Standard errors clustered at the plume level.

baseline soil quality using gridded data on soil clay percentage for Colombia²⁰. All other terms are the same as Equation 3. β_2 is the coefficient of interest; a positive value implies that higher clay content mitigates the adverse effects of glyphosate exposure.

Table 3 presents estimates of β_2 for soil quality. Consistent with the proposed mechanism, higher clay content cushions the damage from glyphosate drift. The point estimate implies that a one percentage point increase in clay composition mitigates the adverse impact of aerial spraying on harvested area by $0.027/0.964 \approx 3\%$ (column 1), with similar moderating effects for output (column 2) and revenue (column 3). We find no heterogeneous effects for NDVI (column 4). These results support the interpretation that clay-rich soils sequester airborne glyphosate before it reaches plant tissues (Wang et al., 2019), reducing the realized exposure of crops to a given level of modeled drift.

In Appendix B.3, we test heterogeneity by crop type and crop diversity, which serve as diagnostic checks that glyphosate is a broad-spectrum herbicide that affects all plants equally. As expected, we find no heterogeneity by crop type or by crop diversity.

5.5 Adaptation: Dynamics, Farm Credit, and GMO Varieties

For adaptive behavior to be observable in our data, three conditions must hold. First, farmers must be repeatedly exposed to glyphosate drift, since transitory damage offers limited incentive to update cropping decisions. Second, farmers must be able to attribute crop damage to herbicide exposure.²¹ And third, farmers must have the financial capacity to finance alternative practices, such as adopting GMO or installing physical barriers. We investigate adaptation by estimating dynamic effects (which would show diminishing damages over time if adaptation occurs), studying borrowing behavior (a proxy for financial capacity), and measuring direct adoption of glyphosate-tolerant GMO varieties.

Figure 3 presents estimates of Equation 3 with lagged exposure terms up to three years. White diamonds report the sum of the contemporaneous and lagged coefficients, capturing the net impact of exposure on harvested area several periods later. If repeated exposure prompts adaptation, then a U-shape should arise as negative effects become muted over time. Instead, we find that contemporaneous damages to off-target crops are persistent. Accumulated damages two years later (Sum “L0-L2”) are nearly equal to the contemporaneous effect. And while the extent of negative spillovers qualitatively increases three years later, we cannot reject the null hypothesis that coefficients are equal since confidence intervals overlap point estimates across all periods. Although this figure is consistent with limited adaptation, we cannot rule out that part of the persistence reflects accumulated soil damage.

Table A6 examines the financial-capacity channel using credit data from EVA. Adaptation through GMO seeds, protective barriers, etc., requires upfront financing, so higher borrowing in exposed municipalities would be consistent with adaptation. Each column reports estimates of Equation 3 where the outcome is credit disbursed to small, medium, and large farm operators in a municipality. While exposure weakly increases total credit (column 1), there are no impacts on individual producer groups (columns 2-4). These results suggest that farmers may be unable to adapt due to limited microcredit access in Colombia.

Lastly, the most direct test of adaptation is to document whether farmers more exposed to glyphosate switch to glyphosate-tolerant GMO varieties. Adoption of GMO crops is the central margin through which commercial farmers respond to herbicide drift (Missirlian, 2024; Young et al., 2023), and a broader literature links GMO adoption to land use, pesticide application, and environmental outcomes (Noack et al., 2024). In Colombia, however, GMO adoption was limited during our study period and concentrated mainly in cotton and maize (Brookes, 2020). We therefore expect limited adaptation through this channel in our setting.

To directly test for GMO adoption, we obtained data on GMO maize and cotton at the

²⁰Data accessible here: <https://essd.copernicus.org/articles/14/4719/2022/>.

²¹Public complaints about aerial spraying in this period are well documented (Defensoría del Pueblo, 2014).

department-year level through a Freedom of Information request (Section 3.1.2). Table A7 reports estimates of Equation 3, where the outcome is the share of harvested area in a department planted with GMO crops. We find no correlation between exposure to aerial spraying and GMO adoption. Because data are at the department-year level, we only have 125 observations and thus cannot exploit the municipality-level wind variation that underpins our main identification strategy. As a result, estimates are correlations and should be interpreted as suggestive evidence of no adaptation. Reassuringly, the results are consistent with our dynamic estimates and evidence on low credit uptake, which also suggest limited adaptation.

5.6 External Validity: Extrapolating to Conventional Applications

Our estimates reflect aerial spraying conducted as a government enforcement policy at much greater intensity than conventional agricultural applications. As noted in Section 2.4, aerial coca eradication applied roughly four to six times the glyphosate active ingredient per hectare used in conventional agriculture (Table 1), in a more concentrated and surfactant-enhanced mixture. Scaling our estimated 13.2% reduction in harvests by the four-to-sixfold difference in per-hectare application rates implies a reduction of roughly 2–3% under conventional application.²² In practice, however, damages may not scale proportionally with dosage because crop responses are often non-linear (Kniss, 2018).

For an independent benchmark, Young et al. (2023) estimate an average soybean yield loss of about 3% from dicamba drift in the United States, where dicamba is sprayed at conventional agronomic doses. The two settings differ along several dimensions, including herbicide chemistry, the affected crop, and application method, but the order of magnitude is informative. Single-digit-percent yield losses under conventional application, compared with the double-digit-percent losses we document under aerial coca eradication, are consistent with our estimates being driven by the high doses under Plan Colombia rather than by glyphosate’s intrinsic phytotoxicity.

Our results should therefore be read as an upper bound on the damages that conventional pesticide drift would produce, and as most directly informative for other settings where governments or large operators apply chemicals at high doses over wide areas, such as military herbicide programs or large-scale aerial fumigation.

²²This is an upper bound on conventional drift damage as coca spraying was also more concentrated and surfactant-enhanced, so its per-kilogram phytotoxicity likely exceeded conventional applications.

6 The Ban on Aerial Spraying

Now that we have established clear negative externalities from aerial spraying, the logical policy prescription is to ban the practice and find alternative ways to curb illegal coca production. The Government of Colombia suspended aerial spraying in May 2015, citing the protection of population health and environmental conservation ([El Consejo Nacional de Estupefacientes, 2015](#)). This section evaluates whether the policy was effective at revitalizing the agricultural sector by enabling damaged crops to recover.

6.1 Policy Evaluation: Difference-in-Differences

We use a difference-in-differences design to study the impact of banning aerial spraying in 2015. Treatment is defined as a continuous variable measuring pre-policy cumulative exposure to aerial spraying, $E_m := \sum_{t=2011}^{2015} E_{mt}$, where E_{mt} is defined in Equation 1. Interacting E_m with a post-policy indicator generates two sources of variation that we exploit for identification: (i) geographic variation in pre-policy exposure to aerial coca spraying across municipalities (Figure A1), and (ii) time variation in crop production before and after the ban. Our difference-in-differences strategy thus compares crop health across previously exposed and less-exposed municipalities, before and after the ban with the following equation:

$$\text{Log } Y_{mt} = \theta \cdot (E_m \cdot \mathbb{1}_{t \geq 2015}) + \mathbf{X}'_{mt} \Omega + \alpha_m + \gamma_t + \epsilon_{mt} \quad (5)$$

where m and t are indexes for municipality and year, respectively. Agricultural outcomes of interest are the same as Equation 3. E_m denotes the treatment, i.e., cumulative exposure of municipality m to aerial spraying during the pre-period. This enters interacted with $\mathbb{1}_{t > 2015}$, a pre-post indicator that switches on in 2015 when aerial spraying was banned. As before, α_m and γ_t are municipality and year fixed effects, respectively.

The coefficient of interest is θ , the impact of banning aerial spraying on crop health in municipalities that were more exposed to glyphosate drift during Plan Colombia compared to those that were less-exposed. Since these control municipalities were minimally exposed in the first place, they have little to recover from and thus form the counterfactual agricultural growth trend. $\theta > 0$ thus implies that agriculture in previously exposed areas grew faster post-ban, diverging from the counterfactual trend and signaling agricultural recovery. Standard errors are clustered by plume (Section 4.1).

Identification of θ requires three assumptions: first, the treatment must be continuous, which is the case since E_m is measured in hectare-equivalents. The second assumption is no anticipation, meaning that farmers do not change their behavior before the ban. The third assumption is parallel trends, which assumes that the evolution of agricultural outcomes

Table 4: Difference in Differences Results

	(1) Area	(2) Output	(3) Revenue	(4) NDVI
Exposure _m × 1 _{t>2015}	0.023*** (0.006)	0.022*** (0.008)	0.006 (0.012)	0.001 (0.000)
Coca + Climate Controls	Yes	Yes	Yes	Yes
Forest Cover	No	No	No	Yes
Municipality FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	8976	8976	8976	8976

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. $Exposure_m$ is pre-policy cumulative exposure. $1_{t>2015}$ is a dummy that switches on after 2015. “Coca Controls” include distance to nearest sprayed area and ha of coca. “Climate Controls” include temperature and rain. Standard errors clustered by plume.

that municipalities with any exposure level, E_m , would have experienced without treatment is the same as the evolution of outcomes that units in the untreated group actually experienced. We test for parallel trends with an event study design next.

6.2 Dynamic Equation

To investigate the dynamic relationship between policy exposure and agricultural production, as well as explore pre-existing agricultural trends in treated municipalities, we present results from an event study version of the main equation:

$$\text{Log } Y_{mt} = \sum_{\tau \in \mathcal{T}^{pre}} \theta_{\tau}(E_m \cdot \gamma_{\tau}) + \sum_{\tau \in \mathcal{T}^{post}} \theta_{\tau}(E_m \cdot \gamma_{\tau}) + \mathbf{X}'_{mt}\Omega + \alpha_m + \gamma_t + \epsilon_{mt} \quad (6)$$

where all terms are the same as in Equation 5. Importantly, $\mathbf{X}'_{mt}$ includes distance from m to the nearest sprayed area, and γ_{τ} is an indicator for year τ . The coefficients of interest are θ_{τ} , where $\theta_{\tau=2014}$ is omitted so that coefficients are measured relative to the year before the ban. If farming activity in treatment and control municipalities were on similar trends prior to the ban, then θ_{τ} should be statistically indistinguishable from zero when $\tau \in \mathcal{T}^{pre}$. When $\tau \in \mathcal{T}^{post}$, the θ_{τ} 's identify the effect of the ban on agricultural outcomes in year τ .

6.3 Impacts of the Ban on Aerial Spraying

6.3.1 Limited Recovery of Agriculture

Estimates of Equation 5 are in Table 4. Column 1 is our preferred specification, where the outcome is log harvested area. We find $\theta > 0$, implying that crop production in municipi-

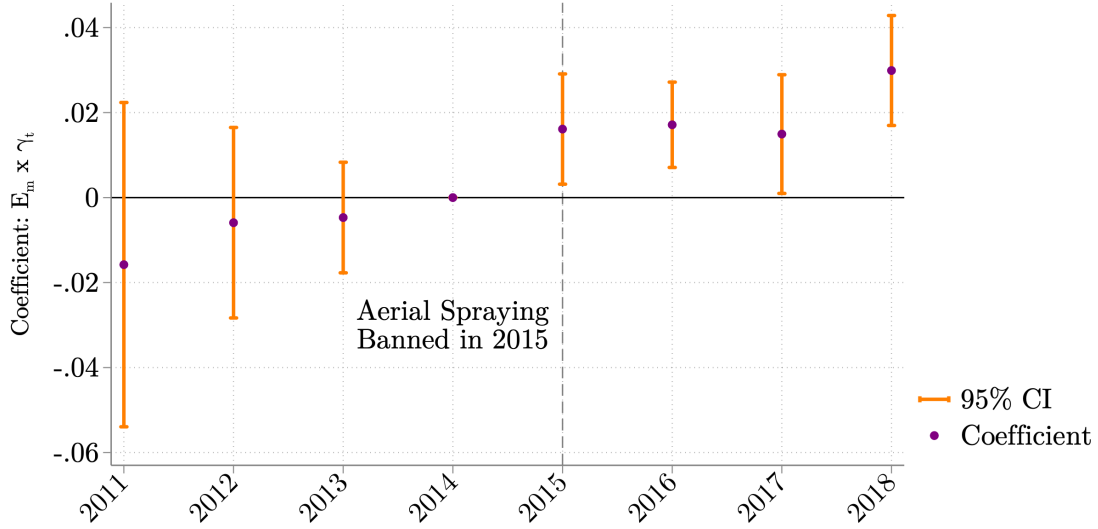


Figure 4: Event Study

Note: Purple circles are coefficients from Equation 6. The omitted period is 2014. Bars are 95% confidence intervals. The regression includes municipality and year fixed effects as well as controls for distance to the nearest sprayed area, coca area, temperature, and rainfall. Standard errors clustered by plume.

palities previously exposed to glyphosate drift begins to rebound once spraying is banned. Point estimates suggest that the ban increased harvested area by 2.3% over the next four years. This limited recovery suggests that much of the damaged cropland did not return to agricultural production after spraying ceased. In Section 6.3.3, we show that much of this land instead transitioned into alternative land uses.

The remaining columns explore sensitivity to alternative outcomes. Column 2 documents robustness to using log crop output (tons). The magnitude of recovery is the same as crop area. By contrast, we find no revenue response (column 3). Given higher output and fixed prices, the attenuation of revenue may be explained by a post-ban shift toward lower-value crops. Column 4 uses NDVI as the outcome and finds no effect. One explanation is that, although we control for forest cover, residual variation in NDVI may still capture other natural vegetation that regenerates more slowly than managed agriculture.

Event study estimates are presented in Figure 4. Prior to the ban on aerial spraying, exposed and non-exposed municipalities are on statistically similar trends and coefficient estimates are near-zero. Once aerial spraying is banned in 2015, the coefficient turns sharply positive and significant for the next four years. Overall, the lack of pre-trends supports the parallel trends assumption. It also supports the no-anticipation assumption, which requires that farmers do not alter their behavior prior to policy.

6.3.2 Robustness of Difference-in-Differences Estimates

Table A8 shows results from robustness tests designed to improve confidence in our difference-in-differences estimates. The first test addresses overlap in the timing of the ban on aerial spraying and the FARC peace deal, a landmark accord that ended decades of armed conflict and aimed to promote rural development in conflict-affected areas of Colombia. Econometrically, this policy overlap is a concern if FARC conflict is a determinant of spraying, which we ruled out in Table A2 (Section 4.2). The overlap is also a concern if FARC conflict is correlated with wind drift, although it is hard to imagine why this would be the case. Nevertheless, we control for $FARC_m \times \mathbb{1}_{t>2015}$ in column 1, where $FARC_m$ is an indicator for whether municipality m was affected by conflict before 2015. The interaction coefficient is statistically indistinguishable from zero, while the coefficient of interest, θ , remains positive, significant, and virtually unchanged in magnitude. The results are stable when $FARC_m$ is measured as the number of pre-policy FARC incidents instead. Our main estimate thus appears uncontaminated by the peace deal timing.

Columns 2-4 test sensitivity to alternative specifications, data sources, and sample restrictions. Column 2 controls for aerial fumigation of legitimate crops such as rice and bananas (see Section 5.3 for data details). Column 3 controls for five meteorological covariates (Section 3.1.3) which partially determine treatment intensity and, potentially, the rate of agricultural recovery. The coefficient remains virtually the same. Lastly, column 4 computes pre-policy exposure, E_m , in Equation 5 with a 30km mask around the point source. This removes treatment variation in the vicinity of the sprayed area which may feature unobserved characteristics that co-determine spraying, thereby helping to isolate wind-driven exposure. The main difference-in-differences coefficient remains stable.

6.3.3 Land Use Change: What Happened to Damaged Cropland?

We have shown that (i) wind drift from glyphosate spraying damaged off-target cropland, and (ii) only a small fraction of these losses were recovered in the four years after spraying was banned. What became of the damaged cropland that failed to recover? We investigate this below by examining land use change in our difference-in-differences framework.

Table A9 reports estimates of θ from Equation 5, where the outcome is the percent of municipality m classified as grassland, shrubland, forest, or bare area in year t . Data are from ESA land cover maps (Section 3.1.2).²³ We find a positive and statistically significant effect for grassland (column 1) and shrubland (column 2), suggesting that municipalities more exposed to glyphosate drift prior to the ban transitioned toward these land types. The transition of previously-exposed cropland into grassland and shrubland is consistent with

²³There are 36 land classes provided. We subsume these into fewer categories using IPCC classifications.

Table 5: Avoided Damages under Different Spraying Regimes

	BAU	Top 50	Top 25	Top 10
Revenue (USD Billions)	62.22439	62.86349	62.86419	62.86552
Avoided Damage (USD Millions)	0	639.0981	639.8025	641.1258

Note: Rows denote total harvest ($TotY^s$) and avoided crop loss ($AvoidedDamage^s$) under each policy scenario. Column 1 is business as usual. Column 2 only sprays areas with above-median coca area. Columns 3-4 target hotspots where coca area is in the top 25th and 10th percentile. Values are means over 1000 bootstrap draws.

several mechanisms. Farmers may abandon agriculture altogether, allowing land to revert to other land covers. On the other hand, farmers may adopt more productive technologies and cultivate less land. Because household-level panel data on labor mobility and agricultural investment are unavailable in Colombia, we cannot distinguish among these channels.

7 Avoidable Damages Under Alternative Policies

In April 2025, Colombia announced plans to reinstate aerial fumigation ([Tactics Institute, 2025](#)). Our findings thus take on new urgency, not only revealing the scale of past spillovers but also informing how spraying policies can be redesigned to reduce collateral damage going forward. In this section, we use our retrospective estimates to simulate how alternative targeting strategies could mitigate spillover crop damage.

Consider three targeting rules that spray only coca hotspots: (i) above the median of the coca area distribution (“Top 50”), (ii) in the top quartile (“Top 25”), or (iii) in the top decile (“Top 10”). Under each scenario, spray campaigns targeting smaller coca plantations are dropped from the dataset, and municipality exposure is recomputed according to Equation 2. Then, for each municipality m and year t , we use our coefficient estimates from Equation 3 to predict crop revenue in (constant) dollars, Y_{mt}^s , under scenario $s \in \{\text{Top 50, Top 25, Top 10}\}$:²⁴

$$\hat{Y}_{mt}^s = \hat{\beta} \cdot E_{mt}^s + \mathbf{X}_{mt}' \hat{\Omega} + \alpha_m + \gamma_t$$

where E_{mt}^s denotes realizations of exposure to spraying under scenario s . These predictions capture the partial-equilibrium effect of alternative coca targeting on spillover damage, holding other factors constant. We then sum local predicted values, $\hat{Y}_{mt}^s$, across all municipalities and years to construct counterfactual measures of national crop revenue under scenario s , denoted by $TotY^s$. Comparing these totals to the in-sample aggregate fitted values

²⁴We use estimates of Equation 3 without taking the log of the outcome so that we can easily aggregate predicted revenue in the next step.

$TotY^{BAU}$ (business as usual) yields a simple metric of damage abated under each scenario:

$$\text{DamageMitigation}^s = \text{TotY}^s - \text{TotY}^{BAU} \quad \text{for } s \in \{\text{Top 50, Top 25, Top 10}\} \quad (7)$$

To account for uncertainty in the predicted values, we bootstrap the prediction and aggregation procedure with 1000 draws and report the mean. Table 5 reports mean aggregate crop revenue and damage mitigation under each scenario. A policy that restricts spraying to coca plantations above the median size avoids \$USD 639 million of spillover crop damage over five years relative to BAU. To contextualize this value against a benchmark, these total savings amount to 3.7% of Colombia’s average agricultural GDP during our study period (World Bank, 2026). A policy that restricts spraying to the largest coca plantations only (top decile) yields additional savings of roughly \$USD 2 million.

Three mechanisms could drive damage mitigation under our proposed targeting rules. First, restricting spraying to large plantations reduces the total number of spray campaigns, lowering overall herbicide release (extensive margin). Second, small coca plots may be more likely to be embedded within mixed agricultural landscapes; excluding them therefore would reduce drift onto nearby legal crops. Lastly, if legitimate crops near small coca plantations are more vulnerable or more valuable, excluding these areas from spraying would yield additional benefits. Although we cannot separately identify these channels, our estimates quantify the total avoided damages implied by simple policy changes.

While aerial fumigation may remain undesirable on other grounds, our simulations suggest that smarter spatial targeting can substantially mitigate its unintended consequences for non-targeted agriculture.

8 Conclusion

This paper showed that geographically targeted enforcement policies can generate large and persistent spatial spillovers. Using Colombia’s aerial coca eradication program as a natural experiment, we document that policies implemented in specific locations can impose substantial costs on non-targeted areas through environmental spillovers. These spillovers are not transitory and persist even after the policy ends.

To quantify spatial spillovers, we combine coca eradication data with a novel atmospheric dispersion model that simulates wind-driven glyphosate drift from sprayed coca fields. This approach yields a plausibly exogenous measure of herbicide exposure across municipalities. Our analysis reveals substantial negative spillovers: municipalities more exposed to glyphosate drift experienced a 13.2% reduction in harvested crop area during the period of active spraying, with negative effects persisting for up to four years after exposure.

In aggregate, our estimates imply that agricultural expansion in Colombia was held back by roughly 12% due to off-target crop losses.

A central feature of our findings is the persistence of off-target damages. We find little evidence of short-run adaptation through credit uptake or GMO crop adoption. Consistent with this, when we study the impact of the 2015 ban on aerial spraying, our difference-in-differences estimates show limited agricultural rebound in previously exposed areas. Instead, we document a transition of damaged cropland into grassland and shrubland, indicating that place-based agricultural policy shocks can lead to longer-run changes in land use rather than temporary production disruptions.

The policy implications of our findings extend beyond Colombia. Geographic targeting is widely used to regulate undesirable activities, from enforcing protected areas to managing pollution and natural resources. Yet environmental externalities, of course, do not always respect geographic boundaries. We document substantial cross-municipality spillovers and show that their effects can last for years after the policy ends, especially when adaptation is limited. Our findings imply that policy evaluation should account not only for direct effects but also for the magnitude and persistence of indirect effects. This is especially important when direct effects are minimal, as in the case of aerial coca eradication ([Mejía, 2016](#); [Mejía and Restrepo, 2016](#)), because spillovers may ultimately dominate the overall policy impact.

At the same time, our findings suggest room for more effective program design. Even when the choice to deploy a place-based instrument is fixed, policymakers retain considerable discretion over which places to target. Our policy simulations show that restricting enforcement to the largest coca hotspots would have avoided roughly 3.7% of agricultural GDP in collateral damage, a meaningful gain from a modest adjustment to policy design.

References

- Abman, Ryan, Eric C Edwards, and Danae Hernandez-Cortes,** “Water, dust, and environmental justice: The case of agricultural water diversions,” *American Journal of Agricultural Economics*, 2024.
- Agostini, Lidiane P, Raquel S Dettogni, Raquel S Dos Reis, Elaine Stur, El-damária VW Dos Santos, Diego P Ventorim, Fernanda M Garcia, Rodolfo C Cardoso, Jones B Graceli, and Iúri D Louro,** “Effects of glyphosate exposure on human health: Insights from epidemiological and in vitro studies,” *Science of the Total Environment*, 2020, 705, 135808.
- Assunção, Juliano, Robert McMillan, Joshua Murphy, and Eduardo Souza-Rodrigues,** “Optimal environmental targeting in the amazon rainforest,” *The Review of Economic Studies*, 2023, 90 (4), 1608–1641.
- Benbrook, C.M,** “Trends in Glyphosate Herbicide Use in the United States and Globally,” *Environmental Sciences Europe*, 2016, 28 (3).
- Bish, Mandy, Eric Oseland, and Kevin Bradley,** “Off-target Pesticide Movement: A Review of Our Current Understanding of Drift Due to Inversions and Secondary Movement,” *Weed Technology*, December 17 2020, 35 (3), 345–356. Published online December 17 2020.
- Blackburn, L. G. and C. Boutin,** “Subtle effects of herbicide use in the context of genetically modified crops: a case study with glyphosate (Roundup),” *Ecotoxicology*, 2003, 12 (1-4), 271–285.
- Blattman, Christopher, Donald P Green, Daniel Ortega, and Santiago Tobón,** “Place-based interventions at scale: The direct and spillover effects of policing and city services on crime,” *Journal of the European Economic Association*, 2021, 19 (4), 2022–2051.
- Boonupara, Thirasant, Patchimaporn Udomkun, Eakalak Khan, and Puangrat Kajitvichyanukul,** “Airborne pesticides from agricultural practices: A critical review of pathways, influencing factors, and human health implications,” *Toxics*, 2023, 11 (10), 858.
- Brookes, G.,** “Genetically modified (GM) crop use in Colombia: farm level economic and environmental contributions,” *GM Crops & Food*, July 2020, 11 (3), 140–153. Epub 2020 Feb 2.
- Butts, Thomas R, Bradley K Fritz, K Badou-Jeremie Kouame, Jason K Norsworthy, L Tom Barber, W Jeremy Ross, Gus M Lorenz, Benjamin C Thrash, Nick R Bateman, and John J**

- Adamczyk**, “Herbicide spray drift from ground and aerial applications: Implications for potential pollinator foraging sources,” *Scientific Reports*, 2022, 12 (1), 18017.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro HC Sant’Anna**, “Difference-in-differences with a continuous treatment,” *NBER WP No. 32117*, 2024.
- Calzada, Joan, Meritxell Gisbert, and Bernard Moscoso**, “The Hidden Cost of Bananas: The Effects of Pesticides on Newborns’ Health,” *Journal of the Association of Environmental and Resource Economists*, November 2023, 10 (6). Forthcoming or In press (if not yet assigned page numbers).
- Camacho, Adriana and Daniel Mejía**, “The Health Consequences of Aerial Spraying Illicit Crops: The Case of Colombia,” *Journal of Health Economics*, July 2017, 54, 147–160. Epub 2017 May 13.
- Cederlund, Harald**, “Effects of spray drift of glyphosate on nontarget terrestrial plants—a critical review,” *Environmental toxicology and chemistry*, 2017, 36 (11), 2879–2886.
- Chih Chang, Feng, Matt F. Simcik, and Paul D. Capel**, “Occurrence and fate of the herbicide glyphosate and its degradate aminomethylphosphonic acid in the atmosphere,” *Environmental Toxicology and Chemistry*, March 2011, 30 (3), 548–555.
- Coinon, Marine**, “Pesticide Externalities and Spatial Coordination Failure in Mixed Farming Landscapes,” 2025. Working Paper.
- Colella, Fabrizio, Rafael Lalive, Seyhun Orcan Sakalli, and Mathias Thoenig**, “Inference with arbitrary clustering,” *IZA Discussion Paper No. 12584*, 2019.
- Conley, Timothy G**, “GMM estimation with cross sectional dependence,” *Journal of econometrics*, 1999, 92 (1), 1–45.
- Defensoría del Pueblo**, “Respuesta a la Proposición 042 de 2014 sobre fumigaciones con glifosato,” Technical Report, Defensoría del Pueblo, Bogotá, Colombia November 2014.
- Delkash-Roudsari, S., A. M. Chicas-Mosier, S. H. Goldansaz, K. Talebi-Jahromi, A. Ashouri, and C. I. Abramson**, “Assessment of Lethal and Sublethal Effects of Imidacloprid, Ethion, and Glyphosate on Aversive Conditioning, Motility, and Lifespan in Honey Bees (*Apis Mellifera* L.),” *Ecotoxicology and Environmental Safety*, 2020, 204, 111108.
- Dewar, A. M., L. A. Haylock, M. J. May, J. Beane, and R. N. Perry**, “Glyphosate Applied to Genetically Modified Herbicide-Tolerant Sugar Beet and ‘Volunteer’ Potatoes Reduces Populations of Potato Cyst Nematodes and the Number and Size of Daughter Tubers,” *Annals of Applied Biology*, 2000, 136, 179–187.

- Dias, Mateus, Rudi Rocha, and Rodrigo R. Soares**, “Down the River: Glyphosate Use in Agriculture and Birth Outcomes of Surrounding Populations,” *The Review of Economic Studies*, November 2023, 90 (6), 2943–2981.
- Didan, Kamel, Armando Barreto Munoz, Ramon Solano, Alfredo Huete et al.**, “MODIS vegetation index user’s guide (MOD13 series),” *University of Arizona: Vegetation Index and Phenology Lab*, 2015, 35, 2–33.
- DIRAN**, “Selección de Preguntas Destacadas y Recurrentes Durante Reuniones Informativas Previas a la Audiencia Pública Ambiental: Modificación del Plan de Manejo Ambiental para el Programa de Erradicación de Cultivos Ilícitos mediante Aspersión Aérea,” Technical Report, Policía Nacional de Colombia, Bogotá, Colombia 2020. Documento utilizado en reuniones informativas del 7, 9 y 11 de mayo de 2020.
- DNE**, “La lucha de Colombia contra las drogas ilícitas: acciones y resultados 2002,” Technical Report, República de Colombia, Ministerio del Interior y de Justicia, Dirección Nacional de Estupefacientes, Bogotá, D.C., Colombia 2003.
- Donkelaar, Aaron Van, Randall V Martin, Michael Brauer, N Christina Hsu, Ralph A Kahn, Robert C Levy, Alexei Lyapustin, Andrew M Sayer, and David M Winker**, “Global estimates of fine particulate matter using a combined geophysical-statistical method with information from satellites, models, and monitors,” *Environmental science & technology*, 2016, 50 (7), 3762–3772.
- Draxler, Roland R and GD Hess**, “An overview of the HYSPLIT.4 modelling system for trajectories,” *Australian meteorological magazine*, 1998, 47 (4), 295–308.
- , **Barbara Stunder, Glenn Rolph, Ariel Stein, and Albion Taylor**, “HYSPLIT4 users’s guide,” Technical Report 2020.
- El Consejo Nacional de Estupefacientes**, “Resolution 6 of 2015,” *Legal Note*, 2015.
- Englander, Gabriel**, “Information and spillovers from targeting policy in peru’s anchoveta fishery,” *American Economic Journal: Economic Policy*, 2023, 15 (4), 390–427.
- Frank, Eyal G.**, “The Economic Impacts of Ecosystem Disruptions: Costs From Substituting Biological Pest Control,” *Science*, 2024, 385 (6713), eadg0344.
- **and Charles A. Taylor**, “The “Golden Age” of Pesticides? Trade-offs of DDT and Health in the US,” 2022. Working Paper.

- Franz, J.E., M.K. Mao, and J.A. Sikorski**, *Glyphosate: A Unique Global Herbicide*, Washington, DC: American Chemical Society, 1997.
- Fritz, B. K., W. C. Hoffmann, and W. E. Bagley**, “Effects of Spray Mixtures on Droplet Size Under Aerial Application Conditions and Implications on Drift,” *Applied Engineering in Agriculture*, 2010, 26 (1), 21–29.
- Gandhi, Kavita, Shaila Khan, Mansi Patrikar, Aishwarya Markad, Niraj Kumar, Aaditee Choudhari, Pallavi Sagar, and Shreya Indurkar**, “Exposure risk and environmental impacts of glyphosate: Highlights on the toxicity of herbicide co-formulants,” *Environmental Challenges*, 2021, 4, 100149.
- Giesy, John P., Sarah Dobson, and Keith R. Solomon**, “Ecotoxicological Risk Assessment for Roundup® Herbicide,” *Reviews of Environmental Contamination and Toxicology*, 2000, 167, 35–120.
- Gray, Wayne B. and Jay P. Shimshack**, “The Effectiveness of Environmental Monitoring and Enforcement: A Review of the Empirical Evidence,” *Review of Environmental Economics and Policy*, Winter 2011, 5 (1), 3–24.
- Hanson, Andrew and Shawn Rohlin**, “Do spatially targeted redevelopment programs spillover?,” *Regional Science and Urban Economics*, 2013, 43 (1), 86–100.
- and —, “A toolkit for evaluating spatially targeted urban redevelopment incentives: Methods, lessons, and best practices,” *Journal of Urban Affairs*, 2021, 43 (5), 618–639.
- Heo, Seonmin Will, Koichiro Ito, and Rao Kotamarthi**, “International Spillover Effects of Air Pollution: Evidence from Mortality and Health Data,” *The Review of Economics and Statistics*, 03 2025, pp. 1–45.
- Hernandez-Cortes, Danae and Kyle C Meng**, “Do environmental markets cause environmental injustice? Evidence from California’s carbon market,” *Journal of Public Economics*, 2023, 217, 104786.
- Hewitt, Andrew J., Keith R. Solomon, and E. J. P. Marshall**, “Spray Droplet Size, Drift Potential, and Risks to Nontarget Organisms from Aerially Applied Glyphosate for Coca Control in Colombia,” *Journal of Toxicology and Environmental Health, Part A*, 2009, 72 (15–16), 921–929.
- Hoffmann, Lars, Gebhard Günther, Dan Li, Olaf Stein, Xue Wu, Sabine Griessbach, Yi Heng, Paul Konopka, Rolf Müller, Bärbel Vogel et al.**, “From ERA-Interim to ERA5:

- the considerable impact of ECMWF's next-generation reanalysis on Lagrangian transport simulations," *Atmospheric Chemistry and Physics*, 2019, 19 (5), 3097–3124.
- Horta-Sáenz, Daniela and Anderson Tami-Patiño**, "Collateral Damage: The Impact of Forced Eradication of Illicit Crops on Human Capital," *Available at SSRN* 4694457, 2024.
- INL**, "Informe sobre temas relacionados con la erradicación aérea de coca ilícita en Colombia," 2002.
- International Agency for Research on Cancer**, "IARC Monographs Volume 112: Evaluation of Five Organophosphate Insecticides and Herbicides," Report 2015. Accessed: 2025-01-22.
- Jordan, T., G. Nice, B. Johnson, and T. Bauman**, "Reducing Spray Drift from Glyphosate and Growth Regulator Herbicides," Technical Report, Purdue University, Department of Botany & Plant Pathology 2009.
- Kniss, Andrew R.**, "Soybean Response to Dicamba: A Meta-Analysis," *Weed Technology*, 2018, 32 (5), 507–512.
- Larsen, Ashley E. et al.**, "Spillover Effects of Organic Agriculture on Pesticide Use on Nearby Fields," *Science*, 2024, 383, eadf2572.
- Marshall, E. J., K. R. Solomon, and G. Carrasquilla**, "Coca (*Erythroxylum coca*) Control is Affected by Glyphosate Formulations and Adjuvants," *Journal of Toxicology and Environmental Health, Part A*, 2009, 72 (15–16), 930–936.
- Meftaul, Islam Md, Kadiyala Venkateswarlu, Prasath Annamalai, Aney Parven, and Mallavarapu Megharaj**, "Glyphosate Use in Urban Landscape Soils: Fate, Distribution, and Potential Human and Environmental Health Risks," *Journal of Environmental Management*, 2021, 292, 112786.
- Mejía, Daniel**, "Plan Colombia: an analysis of effectiveness and costs," *Center for 21st Century Security and Intelligence-Latin America Initiative*, 2016.
- Mejia, Daniel and Pascual Restrepo**, "The economics of the war on illegal drug production and trafficking," *Journal of Economic Behavior & Organization*, 2016, 126, 255–275.
- MinAmbiente and MinSalud**, "Orden décimo quinta del auto 073 de 2014: Parte I Glifosato," 2014. Accessed: 2025-01-27.
- Missirian, Anouch**, "Yes, in Your =Backyard: Forced Technological Adoption and Spatial Externalities," Working Paper 2024.

- Morehouse, John M. and Edward Rubin**, “Downwind and Out: The Strategic Dispersion of Power Plants and Their Pollution,” *Journal of the Association of Environmental and Resource Economists*, 2026, 13 (3), 831–867.
- Myers, John P., Michael N. Antoniou, Bruce Blumberg, and et al.**, “Concerns Over Use of Glyphosate-Based Herbicides and Risks Associated with Exposures: A Consensus Statement,” *Environmental Health*, 2016, 15, 19.
- Neumark, David and Helen Simpson**, “Place-Based Policies,” in Gilles Duranton, J. Vernon Henderson, and William C. Strange, eds., *Handbook of Regional and Urban Economics*, Vol. 5, Elsevier, 2015, pp. 1197–1287.
- New York Times**, “Defying US, Colombia halts aerial spraying of crops used to make cocaine,” *New York Times*, 2015, 14.
- Nivia, Elsa**, “Las fumigaciones sí son peligrosas,” February 2002. Retrieved July 11, 2025.
- Noack, Frederik, Dennis Engist, Josephine Gantois, Vasundhara Gaur, Batoule F. Hyjazie, Ashley Larsen, Leithen K. M’Gonigle, Anouch Missirian, Matin Qaim, Risa D. Sargent, Eduardo Souza-Rodrigues, and Claire Kremen**, “Environmental impacts of genetically modified crops,” *Science*, 2024, 385 (6712).
- OECD**, “OECD Review of Agricultural Policies: Colombia 2015,” *OECD Publishing*, 2015.
- OECD**, “Agricultural Policy Monitoring and Evaluation 2023: Adapting Agriculture to Climate Change,” Technical Report, Paris 2023.
- PAN**, *Glyphosate Monograph*, Penang, Malaysia: Pesticide Action Network Asia and the Pacific, 2016.
- Ravier, Sylvain, Marine Désert, Grégory Gille, Alexandre Armengaud, Henri Wortham, and Étienne Quivet**, “Monitoring of Glyphosate, Glufosinate-ammonium, and (Aminomethyl)phosphonic acid in ambient air of Provence–Alpes–Côte-d’Azur Region, France,” *Atmospheric Environment*, 2019, 204, 59–68.
- Republic of Colombia**, “Case Concerning Aerial Herbicide Spraying (Ecuador v. Colombia): Counter-Memorial of the Republic of Colombia, Volume I,” International Court of Justice March 2010.
- Reyes, Camilo**, “Intervención del Viceministro de Relaciones Exteriores de Colombia en la sesión del Consejo Permanente celebrada el 9 de enero de 2007,” Organización de los Estados Americanos, Centro de Noticias, Discursos January 2007. Publicado el 10 de enero de 2007.

- Reyes, L. C.**, “Estimating the Causal Effect of Forced Eradication on Coca Cultivation in Colombian Municipalities,” *World Development*, 2014, 61, 70–84.
- Reynier, Emmett and Edward Rubin**, “Glyphosate exposure and GM seed rollout unequally reduced perinatal health,” *Proceedings of the National Academy of Sciences*, 2025, 122 (3), e2413013121.
- Rincón-Ruiz, Alexander and Giorgos Kallis**, “Caught in the Middle, Colombia’s War on Drugs and Its Effects on Forest and People,” *Geoforum*, 2013, 46, 60–78.
- Rivas-Garcia, T., A. Espinosa-Calderón, B. Hernández-Vázquez, and R. Schwentesius-Rindermann**, “Overview of Environmental and Health Effects Related to Glyphosate Usage,” *Sustainability*, 2022, 14 (11), 6868.
- Sanderson, J. B., J. A. MacLeod, and J. Kimpinski**, “Glyphosate Application and Timing of Tillage of Red Clover Affects Potato Response to N, soil N Profile, and Root and Soil Nematodes,” *Canadian Journal of Soil Science*, 1999, 79 (1), 65–72.
- Skidmore, Marin Elisabeth, Kaitlyn M Sims, and Holly K Gibbs**, “Agricultural intensification and childhood cancer in Brazil,” *Proceedings of the National Academy of Sciences*, 2023, 120 (45), e2306003120.
- Solomon, Keith R., Arturo Anadón, Antonio L. Cerdeira, Jon Marshall, and Luz-Helena Sanin**, “Environmental and Human Health Assessment of the Aerial Spray Program for Coca and Poppy Control in Colombia,” Technical Report, Inter-American Drug Abuse Control Commission (CICAD), Organization of American States (OAS), Washington, DC 2005.
- Tactics Institute**, “Colombia reinstates Coca fumigation amid U.S. pressure and domestic dilemma,” *Tactics Institute for Security and Counter Terrorism*, 2025.
- Tokatlian, Juan Gabriel**, “Los narcocrátas no le temen al glyphosato,” in “Drogas, dilemas y dogmas: Estados Unidos y la narcocriminalidad organizada en Colombia” Estudios internacionales, Bogotá, Colombia: Tercer Mundo, 1995, chapter 6, pp. 67–73.
- , “Política antidrogas de Estados Unidos y cultivos ilícitos en Colombia: La funesta rutinización de una estrategia desacertada,” *Análisis Político*, 1998, (35), 47–69.
- Townshend, John, Matthew Hansen, Mark Carroll, Charlene DiMiceli, Robert Sohlberg, and Chengquan Huang**, “Annual global automated MODIS vegetation continuous fields

- (MOD44B) at 250 m spatial resolution for data years beginning day 65, 2000-2014, collection 5 percent tree cover, version 6.," Technical Report, University of Maryland, College Park 2017.
- Tzanetou, Evagelia and Helen Karasali**, "Glyphosate Residues in Soil and Air: An Integrated Review," in Dimitrios Kontogiannatos, Anna Kourti, and Kassio Ferreira Mendes, eds., *Pests, Weeds and Diseases in Agricultural Crop and Animal Husbandry Production*, Rijeka: IntechOpen, 2020, chapter 8.
- UNODC**, "World Drug Report 2024," Technical Report, United Nations 2024. United Nations publication.
- **and Government of Colombia**, "Colombia: Monitoreo de cultivos de coca," Technical Report, United Nations Office on Drugs 2011-2015.
- U.S. Department of State**, "Chemicals Used for the Aerial Eradication of Illicit Coca in Colombia and Conditions of Application," Technical Report, Bureau of International Narcotics and Law Enforcement Affairs, U.S. Department of State 2002. Archived content available from January 20, 2009 to January 20, 2017.
- , "Report on Issues Related to the Aerial Eradication of Illicit Coca in Colombia," *Bureau of International Narcotics and Law Enforcement Affairs Report*, 2003.
- van den Berg, F., R. Kubiak, W. G. Benjey, M. S. Majewski, S. R. Yates, G. L. Reeves, J. H. Smelt, and A. M. A. van der Linden**, "Emission of Pesticides into the Air," *Water, Air, & Soil Pollution*, 1999, 115 (1-4), 195-218.
- Vargas, Ricardo, Mara Lucia Guardiola, and Jorge Almansa**, "Fumigación en Colombia Los exitos del desacierto," Technical Report, Accion Andina 2003.
- Vieira, João Pedro, Ricardo Dahis, and Juliano J Assunção**, *The Role of Sanctions and Spillovers in Forest Conservation*, Monash University, Monash Business School, Department of Economics, 2023.
- Wang, Meichen, Asuka A. Orr, Shujun He, Chimeddulam Dalaijamts, Weihsueh A. Chiu, Phanourios Tamamis, and Timothy D. Phillips**, "Montmorillonites can tightly bind glyphosate and paraquat reducing toxin exposures and toxicity," *ACS Omega*, 2019, 4 (18), 17702-17713.
- Wen, Jeff, Sam Heft-Neal, Patrick Baylis, Judson Boomhower, and Marshall Burke**, "Quantifying fire-specific smoke exposure and health impacts," *Proceedings of the National Academy of Sciences*, 2023, 120 (51), e2309325120.

World Bank, “Agriculture, Value Added (Constant 2015 US\$) – Colombia,” <https://data.worldbank.org/indicator/NV.AGR.TOTL.KD?locations=CO> 2026. World Development Indicators, accessed January 24, 2026.

Young, J. S., T. McCarty, S. Lancaster, and M. Bish, “Modeling Soybean Planting Decisions with Network Diffusion: Does Herbicide Drift Affect Farmer Profitability and Seed Selection?,” *Journal of Agricultural and Applied Economics*, 2023, 55 (2), 304–323.

Yuan, Ye, Shuo Li, Tiancong Chen, and Jianlin Ren, “Effects of Ambient Temperature and Humidity on Natural Deposition Characteristics of Airborne Biomass Particles,” *International Journal of Environmental Research and Public Health*, 2023, 20 (3), 1890.

A Online Appendix

A.1 Appendix Tables

Table A1: Summary Statistics

	Obs.	Mean	Std. Dev.
<i>A: Agriculture (2011-2018)</i>			
Planted Area (ha.)	8976	4548.61	6701.46
Harvested Area (ha.)	8976	3879.96	5745.71
Production (tons)	8976	44333.85	182055.90
NDVI	8976	0.69	0.08
<i>B: Aerial Spraying (2011-2015)</i>			
Exposure (ha.)	5610	0.07	0.34
Sprayed Area (ha.)	5610	5639.88	30045.02
<i>C: Covariates (2011-2018)</i>			
Km to nearest sprayed area	5610	107.94	83.36
Coca (=1)	8976	0.18	0.38
Coca Area (ha.)	8976	3314.38	14302.80
Rain (mm)	8976	9.17	6.45
Temperature (° C)	8976	19.87	5.16

Note: Summary statistics of main outcome variables (Panel A), explanatory variables (Panel B), and covariates (Panel C). Aerial spraying data is available until 2015. All other data until 2018.

Table A2: Selection into Exposure

	(1) Exposure	(2) Exposure
Coca Area (t-1)	0.524** (0.197)	0.439** (0.186)
Harvest Area (t-1)	-0.262 (0.187)	-0.228 (0.163)
Revenue (t-1)	-0.005 (0.055)	-0.010 (0.043)
Temperature (t-1)	0.179 (0.419)	-0.023 (0.378)
Rain (t-1)	0.076 (0.078)	0.137** (0.060)
FARC Incidents (t-1)	0.030 (0.036)	-0.008 (0.024)
Mask	None	30km
Municipality FEs	✓	✓
Year FEs	✓	✓
Observations	3366	3366

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. All variables are in standard deviations. In column 1, the outcome is exposure to aerial spraying in time t . In column 2, exposure is measured with a 30km mask. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

Table A3: Placebo Estimates: Impact of Exposure to Non-sprayed Coca Plantations

	(1) Area	(2) Output	(3) Revenue	(4) NDVI
Exposure (std. dev.)	-0.044** (0.019)	-0.042** (0.017)	-0.047*** (0.013)	-0.005** (0.002)
Placebo Exposure (std. dev.)	-0.007 (0.008)	-0.011 (0.007)	-0.008 (0.006)	-0.001 (0.001)
Coca + Climate Controls	Yes	Yes	Yes	Yes
Forest Cover	No	No	No	Yes
Municipality FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	5610	5610	5610	5610

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. Outcomes are in logs. "Exposure" to aerial spraying is measured in hectare-equivalents. "Placebo Exposure" is exposure to *unsprayed* coca plantations. "Coca Controls" include distance to nearest sprayed area and hectares of coca. "Climate Controls" include temperature and rainfall. Both explanatory variables are measured in standard deviations. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

Table A4: Impacts on Ambient Pollution

	(1) log(PM2.5)	(2) log(SO2)	(3) log(NO2)
Exposure (ha.)	0.003 (0.009)	0.009 (0.008)	0.004 (0.007)
Coca + Climate Controls	Yes	Yes	Yes
Municipality FEs	✓	✓	✓
Year FEs	✓	✓	✓
Observations	5610	5610	5600

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. "Exposure" to aerial spraying measured in hectare-equivalents. "Coca Controls" include distance to nearest sprayed area and hectares of coca. "Climate Controls" include temperature and rainfall. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

Table A5: Robustness Checks – TWFE Estimates

	(1) Area	(2) Area	(3) Area	(4) Area	(5) Area	(6) Crop %	(7) Area	(8) Area
Exposure (ha.)	-0.132** (0.057)	-0.131** (0.057)	-0.131** (0.056)	-0.093* (0.050)	-0.091*** (0.030)	-0.087*** (0.028)	-0.254** (0.097)	-0.341* (0.174)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Protected Area	Yes	No	No	No	No	No	No	No
Legal Spray	No	Yes	No	No	No	No	No	No
Meteorology	No	No	Yes	No	No	No	No	No
Municipality FEs	✓	✓	✓	✓	✓	✓	✓	✓
Department × Year FEs				✓				
Year FEs	✓	✓	✓		✓	✓	✓	✓
Linear Trend	No	No	No	No	Yes	No	No	No
Data Source	EVA	EVA	EVA	EVA	EVA	ESA	EVA	EVA
Mask	No	No	No	No	No	No	30km	No
Sample Restricted	No	No	No	No	No	No	No	Yes
Observations	5610	5610	5610	5605	5610	5610	5610	4480

Note: Data are at the municipality-year level. “Exposure” to aerial spraying measured in hectare-equivalents. The outcome in columns 1-5 and 7-8 are log harvested area. The outcome in column 6 is cropland percent. Column 1 controls for protected area share. Column 2 controls for legal crop spraying. Column 3 controls for energy flux, planetary boundary layer, velocity, humidity, and surface roughness. Column 4 includes department-year fixed effects. Column 5 adds a linear municipality trend. Column 6 uses alternative crop data from ESA. Column 7 adds a 30km mask around the point source before calculating exposure. Column 8 drops sprayed municipalities from the sample. All specifications control for distance to nearest sprayed area, ha of coca, temperature, and rain. Standard errors clustered by plume.

Table A6: Adaptation: Farm Credit

	(1) All Farms	(2) Small Farms	(3) Medium Farms	(4) Large Farms
Exposure (ha.)	0.315* (0.165)	-0.164 (0.115)	0.058 (0.070)	-0.088 (0.143)
Coca + Climate Controls	Yes	Yes	Yes	Yes
Municipality FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	5610	5610	5610	5610

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. Outcome is $\log(\text{farm credit} + 1)$ for different farm operations. “Exposure” to aerial spraying measured in hectare-equivalents. “Coca Controls” include distance to nearest sprayed area and ha of coca. “Climate Controls” include temperature, and rain. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

Table A7: Adaptation: GMO Crop Adoption

	(1) GMO Cotton	(2) GMO Maize
Exposure (ha.)	-0.107 (0.076)	-0.447 (0.264)
Coca + Climate Controls	Yes	Yes
Department FEs	✓	✓
Year FEs	✓	✓
Observations	125	125

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the department-year level. The outcome is share of harvested area planted with GMO cotton or maize. “Exposure” to aerial spraying is measured in hectare-equivalents. “Coca Controls” include distance to nearest sprayed area (department-year mean) and ha of coca (department-year total). “Climate Controls” include temperature, and rain (department means). All specifications include department and year fixed effects. Standard errors clustered at the department level.

Table A8: Robustness: Difference-in-Differences Estimates

	(1) Area	(2) Area	(3) Area	(4) Area
Exposure _{<i>m</i>} × $\mathbb{1}_{t>2015}$	0.021*** (0.007)	0.022*** (0.006)	0.021*** (0.006)	0.038*** (0.012)
FARC _{<i>m</i>} × $\mathbb{1}_{t>2015}$	0.026 (0.058)			
Controls	Yes	Yes	Yes	Yes
Legal Spray	No	Yes	No	No
Meteorology	No	No	Yes	No
Municipality FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Data Source	EVA	EVA	EVA	EVA
Mask	No	No	No	30km
Observations	8976	8976	8976	8976

* $p < .1$, ** $p < .05$, *** $p < .01$. Note: Data are at the municipality-year level. Exposure_{*m*} is the pre-policy total exposure of municipality *m*. $\mathbb{1}_{t>2015}$ is a dummy that switches on in 2015. FARC_{*m*} is a dummy for whether *m* experienced FARC-related conflict in the pre-period. Column 2 controls for legal crop spraying. Column 3 controls for energy flux, planetary boundary layer, velocity, humidity, and surface roughness. Column 4 computes Exposure_{*m*} with a 30km mask around the point source. All specifications control for distance to nearest sprayed area, ha of coca, temperature, and rain. Errors clustered by plume.

Table A9: Difference in Differences Results: Land Use Transitions

	(1) Grassland	(2) Shrubland	(3) Forest	(4) Bare
Exposure _{<i>m</i>} × 1 _{<i>t</i>>2015}	0.003** (0.001)	0.006* (0.003)	-0.016 (0.011)	-0.000 (0.000)
Coca + Climate Controls	Yes	Yes	Yes	Yes
Municipality FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	8976	8976	8976	8976

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. Outcomes are the percent of land under each land use type. *Exposure_{*m*}* is the pre-policy total exposure of municipality *m*. 1_{*t*>2015} is a time dummy that switches on in 2015. “Coca Controls” include distance to nearest sprayed area and hectares of coca. “Climate Controls” include temperature and rainfall. Standard errors clustered by plume.

A.2 Appendix Figures

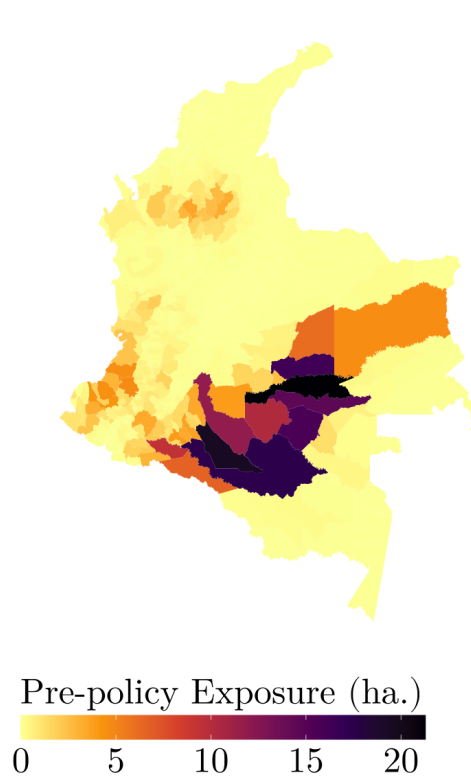


Figure A1: Treatment Distribution: Pre-policy Exposure

Note: Treatment is cumulative exposure during the pre-policy period.

B Supplementary Materials

B.1 Additional HYSPLIT Model Details

Parameterization We run the HYSPLIT model using the `splitr` package in R. We use the dispersion setting, which releases 60,480 tracer particles (normalized to one when we measure exposure; Section 3.2.1) and tracks their movement over time and space as they move with prevailing winds. This is in contrast to the trajectory setting, which traces a single path.

We make three modeling choices that govern how particles disperse. First, particles are released at a height of 50 meters, consistent with documented aircraft spraying altitudes under Plan Colombia (U.S. Department of State, 2003). Second, we do not impose an explicit distance threshold on particle movement due to a lack of evidence from Colombia on the geographic extent of deposition.²⁵ We instead allow distance decay to emerge endogenously through the atmospheric dispersion process built into HYSPLIT. Third, particles are tracked for 24 hours following the spray. This establishes an upper bound on potential atmospheric transport, after which particle trajectories are truncated. We discuss implications of these modeling choices for measurement error below.

Note that our objective with these modeling choices is not to recover the exact concentration or deposition of glyphosate droplets, but rather to identify which municipalities are exposed to glyphosate drift through prevailing wind patterns. HYSPLIT is simply used to trace the movement and dispersion of airborne tracer particles through the atmosphere. The resulting exposure measure is then constructed from the spatial distribution of these particles according to the procedure described in Section 3.2.1.

Particle Deposition We wish to clarify that our HYSPLIT implementation models atmospheric transport rather than the precise deposition of glyphosate onto crop surfaces. This is because data on glyphosate release rates, deposition velocities, and atmospheric concentrations are unavailable for Plan Colombia. Nevertheless, our approach still captures the key implication of deposition: exposure attenuates with distance. While deposition and dispersion are distinct physical processes, both reduce particle concentrations as distance from the source increases. Deposition does so by removing particles from the plume, whereas dispersion dilutes them across a larger volume of air. HYSPLIT models this dispersion directly by enabling turbulence to spread the plume horizontally and vertically (Draxler and Hess, 1998; Draxler et al., 2020). As the plume expands, a fixed quantity of particles is dispersed over an increasingly large volume of air, lowering particle concentrations with distance to the

²⁵An exception is Hewitt et al. (2009), which finds that most glyphosate deposition occurs within a few hundred meters of the spray zone.

source. Thus, even without explicitly modeling deposition, we preserve the central feature that exposure weakens as particles travel farther from the source.

Measurement Error The absence of glyphosate-specific deposition parameters primarily introduces measurement error rather than bias. Our exposure measure is designed to capture the likelihood that tracer particles released from a sprayed area reaches a given location, not the exact quantity of glyphosate deposited there. If glyphosate settles more rapidly than the tracer particles, some distant municipalities may be assigned positive exposure despite receiving no deposition. This would lead to attenuation bias, as exposure would be measured with noise but remain uncorrelated with unobserved determinants of agricultural outcomes (see Section 4.2). To the extent that deposition decays with distance more rapidly than our simulated particle trajectories, the estimated coefficients would represent a lower bound on the effects of actual glyphosate exposure.

B.2 Additional Robustness Tests

A key concern with continuous treatment is that treatment intensity may itself be an endogenous choice by units, which can bias the TWFE estimand (Callaway et al., 2024). In our context, exposure intensity is not chosen by municipalities or farmers. Conditional on the decision of the government to spray a coca field (which we control for), municipalities' exposure to the resulting glyphosate drift is determined by atmospheric processes. Municipalities thus do not select into higher or lower doses, nor can they adjust behavior to influence exposure. Still, Table A10 assesses whether our results could be driven by comparisons across different exposure intensities, as cautioned by Callaway et al. (2024). The resulting effects are qualitatively consistent with our baseline estimates.

Table A10: High- and Low-Exposure Effects Relative to Zero Exposure

	(1) High vs 0	(2) Low vs 0
High Exposure	-0.141 (0.091)	
Low Exposure		-0.038** (0.017)
Coca Controls	Yes	Yes
Climate Controls	Yes	Yes
Municipality FEs	✓	✓
Year FEs	✓	✓
Observations	3682	3689

Note: * $p < .1$, ** $p < .05$, *** $p < .01$. High and low exposure regimes are defined based on discretized values of the exposure measure. Column 1 compares municipalities experiencing high exposure to never-exposed municipalities, while Column 2 compares low exposure to never-exposed municipalities. Never-exposed municipalities are defined as municipality-year observations with zero wind-driven exposure. All specifications include municipality and year fixed effects, coca-related controls, and climate controls. Standard errors are clustered at the municipality level.

Table A11 reports estimates from alternative methods of inference. Recall that our baseline estimates cluster at the plume level using machine learning to demarcate 50 plumes (Section 4.2). While also arbitrary, column 1 tests robustness to demarcating 30 plumes instead. Column 2 clusters by municipality, the same level as the treatment. Columns 3-5 investigate spatial correlation more systematically by estimating Conley (1999) standard errors for kernel cut-off distances ranging from 200km to 500km.²⁶ The statistical precision of our estimates is robust to these alternative methods of accounting for spatial correlation.

²⁶We use the Colella et al. (2019) implementation with the acreg Stata package.

Table A11: Robustness of TWFE Estimates: Standard Errors

	Standard Error Boundary		Conley Spatial Error Cutoff		
	(1) Plume	(2) Municipality	(3) 200km	(4) 300km	(5) 500km
Exposure (ha.)	-0.132** (0.057)	-0.132** (0.059)	-0.132*** (0.048)	-0.132** (0.058)	-0.132*** (0.051)
Controls	Yes	Yes	Yes	Yes	Yes
Municipality FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Clustering	Plume (30)	Municipality	Conley	Conley	Conley
Observations	5610	5610	5610	5610	5610

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. Coefficient estimates and standard errors are from baseline equation (Equation 3) with alternative error clustering. Column 1 clusters at the plume level with 30 demarcated plumes. Column 2 clusters at the municipality level. Columns 3-6 implement [Conley \(1999\)](#) standard errors for four different values of the kernel cut off distance (in km)

B.3 Heterogeneity by Crop Type

The broad-spectrum nature of glyphosate predicts limited heterogeneity by crop diversity or crop type since the herbicide affects nearly all (non-tolerant) crops through the same biological pathway. Table A12 reports estimates of β_2 from Equation 4, where S_m is crop diversity in 2011. We use the EVA data and compute diversity using the Shannon Index,

$$S_m = - \sum_{c=1}^{C_m} p_{cm} \ln(p_{cm})$$

where p_{cm} is the share of total planted area in municipality m devoted to crop c , and C_m is the number of crop species grown in the municipality. The Shannon Index increases with both the number of crop species cultivated (richness) and the evenness with which planted area is distributed across crops. The interaction coefficients are statistically indistinguishable from zero for all outcomes.

Table A13 confirms this lack of heterogeneity by replacing S_m with species richness, an alternative diversity measure that simply counts the number of unique crops without area weighting. We again find no evidence of treatment heterogeneity by crop diversity. Taken together, these results suggest that crop diversification offers little protection against a non-selective herbicide that affects most crops similarly.

The broad-spectrum nature of glyphosate implies that we should also find no heterogeneity by crop type. Although annuals and perennials differ in lifecycle and regenerative capacity, glyphosate disrupts the same biological pathway in both. Table A14 presents estimates of Equation 3 separately for nine types of annual crops, with log harvested area as the outcome. Damages from glyphosate spillovers are most pronounced for oil seeds, fibers, and an “other” category defined by EVA. Similarly, Table A15 shows separate estimates for various perennial crop types, where negative spillovers are strongest for plantains and tree crops. However, when crops are combined into separate annual and perennial totals (column 1 in both tables), effects are statistically insignificant. The absence of a clear annual-versus-perennial difference is again consistent with the broad-spectrum mechanism.

Table A12: Heterogeneity by Crop Diversity (Shannon Index)

	(1) Area	(2) Output	(3) Revenue	(4) NDVI
Exposure (ha.)	-0.234 (0.255)	-0.246 (0.222)	-0.218 (0.177)	0.013 (0.016)
Exposure (ha.) $\times$ Shannon Index	0.049 (0.135)	0.058 (0.120)	0.037 (0.089)	-0.013 (0.008)
Coca + Climate Controls	Yes	Yes	Yes	Yes
Forest Cover	No	No	No	Yes
Municipality FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	5610	5610	5610	5610

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. “Exposure” to aerial spraying is measured in hectare-equivalents. “Coca Controls” include distance to nearest sprayed area and hectares of coca. “Climate Controls” include temperature, and rainfall. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

Table A13: Heterogeneity by Crop Diversity (Species Richness)

	(1) Area	(2) Output	(3) Revenue	(4) NDVI
Exposure (ha.)	-0.260 (0.155)	-0.224 (0.144)	-0.264** (0.103)	0.006 (0.012)
Exposure (ha.) $\times$ Number of Crops	0.013 (0.018)	0.010 (0.018)	0.013 (0.011)	-0.002 (0.001)
Coca + Climate Controls	Yes	Yes	Yes	Yes
Forest Cover	No	No	No	Yes
Municipality FEs	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓
Observations	5610	5610	5610	5610

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. “Exposure” to aerial spraying measured in hectare-equivalents. “Coca Controls” include distance to nearest sprayed area and hectares of coca. “Climate Controls” include temperature, and rainfall. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

Table A14: Impacts of Exposure on Annual Crops

	(1) All	(2) Oilseeds	(3) Fibers	(4) Cereals	(5) Flower	(6) Veggies	(7) Legumes	(8) Spices	(9) Other
Exposure (ha.)	-0.150 (0.110)	-0.281** (0.129)	-0.023* (0.012)	-0.085 (0.099)	0.001 (0.005)	0.004 (0.058)	-0.161 (0.177)	0.025 (0.019)	-0.016*** (0.005)
Coca + Climate Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓	✓	✓	✓
Observations	5478	5478	5478	5478	5478	5478	5478	5478	5478

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. “Exposure” to aerial spraying measured in hectare-equivalents. “Coca Controls” include distance to nearest sprayed area and hectares of coca. “Climate Controls” include temperature, and rainfall. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.

Table A15: Impacts of Exposure on Perennial Crops

	(1) All	(2) Fruits	(3) Plantains	(4) Tree Crops	(5) Other
Exposure (ha.)	-0.058 (0.041)	0.037 (0.083)	-0.078** (0.034)	-0.297* (0.158)	-0.116 (0.082)
Coca + Climate Controls	Yes	Yes	Yes	Yes	Yes
Municipality FEs	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓
Observations	5478	5478	5478	5478	5478

* $p < .1$, ** $p < .05$, *** $p < .01$. Data are at the municipality-year level. “Exposure” to aerial spraying measured in hectare-equivalents. “Coca Controls” include distance to nearest sprayed area and hectares of coca. “Climate Controls” include temperature, and rainfall. All specifications include municipality and year fixed effects. Standard errors clustered at the plume level.